

2 Interaction of light with electrons

2.1 Introduction

In the previous chapter of this book, we have described transport properties of electrons in solids, in particular at mesoscopic length scales. The quantum mechanical phase of the electron wave function, which is decisive for the interference of electrons, was crucial for the measurable conductivity in different types of devices. Hence, quantum mechanics was central. However, we mostly did not solve the Schrödinger or Dirac equation explicitly. Instead, we have built on a more intuitive description of general wave properties in order to explain the different experimentally relevant effects.

This points to a very general approach in experimental solid state physics, justified by the impossibility of solving the fundamental equations such as the Schrödinger equation rigorously for the many degrees of freedom present in a solid. Instead, one needs to operate with adequate approximations, often chosen by intuition or guided by experimental results and justified in detail later. Thus, one often switches between different model languages, appropriate for simple intuition, but consistent with respect to the subsequent detailed justification.

In the present chapter of the book, which deals with the optical properties of solids, we will switch even more extensively between different model languages employing mechanical, electrodynamical, and quantum mechanical explanations. Concretely, we will describe the interaction of electrons with light, which, e.g., determines how we see solids and provides us with means to distinguish between different solids. The detailed understanding of the resulting optical properties has enabled many technical applications of tremendous importance in daily life such as lasers, photodiodes, and solar cells. The interaction of light with electrons is also of fundamental interest as it gives us access to complex many-body interactions of the electrons in solids, whose understanding is still a central challenge in solid state physics (Chapters 4 and 5). While some of these issues can be understood by referring to classical, electrodynamic models describing light as an electromagnetic field and the electrons as classical particles, some of them require treating either the electrons or light or both quantum mechanically.

In order to deal with these different levels of complexity, we will first review some of the electrodynamics of light in solids as well as quantum mechanical descriptions of the transition between different energy levels of an electron due to an electromagnetic field. We will then focus on a special type of optical excitations in solids, namely interband transitions, in which an optical transition of an electron from one band to another occurs. There, we treat the electrons quantum mechanically, namely as Bloch states located in electron bands, but introduce the action of the light as purely provid-

ing the energy for the transition, and keep their quantum mechanical photon character in mind. The interband transitions are of particular importance for the optical properties of semiconductors and are the base for several optoelectronic devices, namely photodiodes, photovoltaic solar cells, light-emitting diodes and semiconductor lasers. After discussing the interband transitions, we will treat the way in which the resulting excited electron and hole can be bound to each other in a state called an exciton. We will see how interband electron-hole excitations can, in turn, emit light, a light emission called luminescence. Finally, we will deal with the free-electron currents in metals and doped semiconductors that give rise to their characteristic reflection spectra.

Along the way, we will familiarize ourselves with some of the most common spectroscopic methods used to measure the optical properties of solids, namely absorption spectroscopy, photoluminescence spectroscopy and reflection spectroscopy.

2.2 Fundamentals of light-matter interaction

In this section, we will summarize some basic aspects of light-matter interaction that are important for treating the optical properties of solids. The light-matter interaction will be considered at different formal levels:

- exclusively using methods from classical physics,
- semi-classically, i.e., matter will be treated quantum mechanically and the light field classically,
- quantum optically, i.e., also the light field is treated quantum physically.

Each level will be chosen according to what is required in order to describe the physical situation correctly and to be as transparent as possible at the same time.

We start by briefly reviewing what should be known a priori from introductory lecture courses or textbooks, namely the central macroscopic terms which describe the light-solid interaction:

Propagation, i.e., the process of light traveling in a medium, usually at a phase velocity lower than in vacuum,

light scattering, i.e., a process in which incident light changes its propagation direction due to an interaction with an optical inhomogeneity in its propagation path,

reflection, i.e., a special case of light scattering in which a smooth solid surface scatters an incident light beam in a single direction into the half-space from which it was incident,

refraction, i.e., a special case of light scattering in which a smooth solid surface transmits an incident light beam in a single direction different from the direction of incidence,

absorption, i.e., an uptake of light energy by the medium without coherent re-emission of this energy at the same frequency.

Microscopically, all of these phenomena result from the interaction of light with electrons in the medium. The high-frequency oscillations of the electromagnetic field (in the visible, near-infrared and ultraviolet frequency ranges) drive periodic motions of these electrons. The resulting charge oscillations lead to the emission of light at the same frequencies and, in non-transparent media, also result in the generation of heat.

2.2.1 Light waves in vacuum

In this section, the properties of light without interaction with matter, i.e., in vacuum, will be briefly summarized. This is needed for an understanding of the following Section 2.2.2.

Light can be described as an electromagnetic wave. Its wave nature is concluded from the characteristic wave behavior that it shows in experiments (propagation, refraction, dispersion, interference, diffraction). Its electromagnetic nature can be concluded from the fact that it is generated by rapidly oscillating charges and exerts forces on charges and currents. Visible light differs from other electromagnetic waves by its wavelength, which ranges from approximately 390 nm to approximately 780 nm, i.e., from the color blue via green, yellow and orange to red. At shorter wavelengths, we have the ultraviolet (UV) regime and at longer wavelengths, the infrared (IR) regime. The term light is also often used for radiation in these wavelength ranges. Light from natural light sources such as incandescent lamps can have many different wavelengths. If it has only one wavelength, e.g., generated by a laser, it is called monochromatic. The spectral distribution of light can be determined experimentally by analyzing it in a grating spectrometer and measuring the light power at the individual wavelengths with a detector. Light waves can theoretically be described using classical electrodynamics. In this approach, light waves represent oscillations of the $\vec{B}$ and $\vec{E}$ field which travel in space, in a process called wave propagation. They obey the electromagnetic wave equation

$$\nabla^2 \vec{E} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{and} \quad \nabla^2 \vec{B} = \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2} \quad (2.1)$$

for the electric field $\vec{E}$ and the magnetic induction $\vec{B}$. ∇ is the Nabla operator and c the phase velocity of the waves. In vacuum, it has the value $c_0 = 3.00 \times 10^8 \text{ m s}^{-1}$. One class of solutions to the wave equation are plane harmonic waves of the form

$$\vec{E}(\vec{r}, t) = \vec{E}_0 \cos(\vec{k} \cdot \vec{r} - \omega t), \quad (2.2)$$

with an analogous form for $\vec{B}$. Here, the wave vector $\vec{k}$ describes the propagation direction of the wave. Its magnitude is connected to the wavelength λ via $|\vec{k}| = 2\pi/\lambda$. Here, ω is the angular frequency. The direction of the amplitude vector $\vec{E}_0$ is the direction of the electric field of the wave, the so-called polarization direction. The magnetic field

$\vec{B}$ obeys the relation

$$\vec{B} = \frac{1}{\omega} (\vec{k} \times \vec{E}) , \quad (2.3)$$

i.e., $\vec{B}$, $\vec{E}$ and the propagation direction of the wave along $\vec{k}$ are pairwise perpendicular to each other. The plane that is spanned by $\vec{E}_0$ and the propagation direction is called the polarization plane. If the polarization plane of the wave remains the same independently of time, the wave is called linearly polarized. For instance, the light from many laser sources is linearly polarized. Light from natural light sources such as the sun or glowing bodies is unpolarized, i.e., this light consists of a superposition of many wave trains with statistically distributed polarization directions. The argument $\vec{k} \cdot \vec{r} - \omega t$ of the cosine function (eq. (2.2)) is called the phase of the wave. All space points of equal phase at a fixed time t are part of a phase front or phase plane. The velocity at which a phase front travels through space is given by the phase velocity of the light wave

$$c = \frac{\lambda}{T_\omega} = \frac{\omega}{k} , \quad (2.4)$$

where T_ω is the time period of the oscillation. The relation derived from this expression,

$$\omega(k) = c \cdot k , \quad (2.5)$$

is often called the dispersion relation of the light wave. If two waves exhibit a spatially or temporally constant difference between their phases, they are called coherent with each other. We define the intensity or energy flow density I_p as the power P transported by the light wave through a unit area A perpendicular to its propagation direction:

$$I_p = \frac{P}{A} . \quad (2.6)$$

The intensity of a plane light wave at a given time t and position $\vec{r}$ can be calculated using eq. (2.2), yielding

$$I_p(\vec{r}, t) = c \varepsilon_0 \vec{E}^2(\vec{r}, t) = c \varepsilon_0 \vec{E}_0^2 \cos^2(\vec{k} \cdot \vec{r} - \omega t) , \quad (2.7)$$

where ε_0 is the vacuum permittivity. Integrating over a time period T_ω of the oscillation and dividing by T_ω gives the time-averaged intensity

$$\bar{I}_p = \frac{c}{2} \varepsilon_0 \vec{E}_0^2 . \quad (2.8)$$

This relates the measurable physical quantity $\bar{I}_p$ to the amplitude $\vec{E}_0$ of the electric field from eq. (2.2). An alternative representation of the electric field is the complex notation

$$\tilde{\vec{E}}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} , \quad (2.9)$$

for which

$$\vec{E}(\vec{r}, t) = \text{Re}[\tilde{\vec{E}}(\vec{r}, t)] = \text{Re}[\vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}] , \quad (2.10)$$

holds. From the complex representation, we thus get back the form of the electric field in eq. (2.2). The complex notation is helpful because in many cases, it allows one to shorten calculations considerably as compared to approaches based on eq. (2.2).

We emphasize that this classical description neglects the corpuscular character of light as described by photons. Hence, it is only applicable at sufficiently high intensities.

2.2.2 Light waves in media

In the previous section, we have dealt with the propagation of light in vacuum, which is an environment that exerts no back effect on the light waves. This applies no longer to the light propagation in media, because the light field causes charge oscillations in the atoms, which can take up energy from the wave, resulting in its gradual attenuation. Moreover, the atoms can, in turn, emit electromagnetic waves, thus giving rise to new contributions to the light field. The medium thus changes the light wave during its propagation. This applies most strongly to condensed matter, which differs from gaseous media by its high number density of atoms. In order to build the basis for the following sections, we will have a closer look at light-matter interaction in media and discuss how one can formally describe the changes experienced by light waves in the course of this interaction.

2.2.2.1 Light absorption and propagation in media

The alternating electric field $\vec{E}(\vec{r}, t)$ can excite forced oscillations of an electron in an atom at the position $\vec{r}$. From a quantum mechanical point of view, such an oscillation consists of a periodic displacement of the probability density of the electron relative to the positively charged atom core. The cause of this displacement is the force $-e\vec{E}(\vec{r}, t)$ exerted on the electron by the alternating electric field of the light wave. For many purposes, it has proven useful to consider the oscillation not from a quantum mechanical perspective but rather classically as a displacement $\vec{x}(t)$ of a point-like electron (Fig. 2.1). The displacement can be treated using the concepts of mechanics and electrodynamics. Moreover, we assume the restoring force exerted on the electron by the atom core is, in linear approximation, proportional to the displacement $\vec{x}(t)$. The system is thus assumed to behave as if the electron and the atom core were connected by a spring. Together, these assumptions are called the Lorentz oscillator model and the system consisting of electron, atom core, and spring is called a Lorentz oscillator. Despite its simplicity, this model yields essential insights into the light matter interaction.

In the Lorentz model, we describe the dynamics of an electron of mass m_e in an atom under the influence of light through an equation of motion for a forced oscillation:

$$m_e \ddot{\vec{x}} + m_e \gamma \dot{\vec{x}} + m_e \omega_0^2 \vec{x} = -e \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}. \quad (2.11)$$

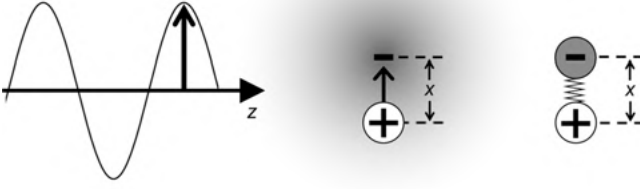


Fig. 2.1: Charge oscillation in an atom under the effect of the alternating electric field of a light wave (left). Center: schematic drawing of the instantaneous displacement of the electronic probability density with respect to the atom core at a given moment in time. Right: Simplified model for the situation depicted in the middle, the so-called Lorentz oscillator. The negative electron charge has its equilibrium position at the position of the positive ion core and is attracted towards the core as if pulled by a spring attached to the electron, which is extended by x .

Here, γ describes a possible damping of the oscillation due to energy losses (for example, due to light emission or collisions of atoms with each other), and ω_0 is the eigenfrequency of the undamped oscillation of the electron. The term on the right side of the equation represents a time-dependent force which drives the oscillation with $\vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$ being the electric field in the complex notation (eq. (2.9)). Recall that $\vec{r}$ is the position of the atom while $\vec{x}$ describes the displacement of the electron charge with respect to the atom core. The complex-valued solution to the differential equation (2.11) for the displacement of the electron has the form

$$\tilde{\vec{x}}(t) = \vec{x}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t + \varphi)}, \quad (2.12)$$

where $\vec{r}$ is again the position of the atom,

$$\vec{x}_0 = \frac{-e\vec{E}_0}{m_e} \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega^2)}} \quad (2.13)$$

is the (real-valued) amplitude of the electron oscillation and φ is the phase delay of the displacement relative to the external force $-e\vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$. For φ we obtain

$$\tan \varphi = \frac{\gamma\omega}{\omega_0^2 - \omega^2}. \quad (2.14)$$

For not too strong damping, the resonance frequency, i.e., the frequency of the exciting light field for which the amplitude becomes maximal (Fig. 2.2(a)), agrees well with the eigenfrequency ω_0 . The displacement $\vec{x}(t)$ of the electron is obtained as the real part of the complex-valued function $\tilde{\vec{x}}(t)$ given by eqs. (2.12), (2.13), and (2.14). We see from eq. (2.12) that the time dependence of the displacement for a given position $\vec{r}$ of the atom represents a harmonic oscillation. Hence the Lorentz oscillator

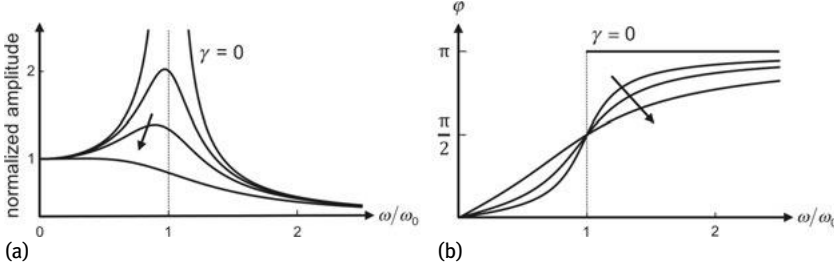


Fig. 2.2: (a) Amplitude of a Lorentz oscillator plotted versus the driving frequency ω for different dampings γ , which increase in the direction of the arrow. The amplitude is normalized to its value at zero frequency (static displacement), and the frequency to the eigenfrequency ω_0 . In the undamped case, the resonance frequency is equal to ω_0 . It decreases with increased damping. (b) Phase shift φ of the displacement relative to the phase of the external force for the same dampings γ as in (a). As in (a), the arrow shows the direction of increased γ .

model describes in a simple way, via eqs. (2.13) and (2.14), the frequency dependence of the electron oscillation. How this frequency dependence is reflected in measurable macroscopic quantities will be considered in the next subsection.

The oscillation energy of the excited electron originates from the light wave. It may lose that energy through the following processes:

- Via collisions (for example with other atoms) or – provided that the atom is part of a molecule – via coupling of the electron oscillation to mechanical oscillations of the molecule. In these processes, the oscillation energy is ultimately transformed into heat. Such an energy uptake with subsequent complete or partial transformation into heat is called absorption. For $\omega \approx \omega_0$, we have resonant absorption.
- Via emission of the energy taken up from the light wave by the atom. The oscillating electron and the corresponding atom core form an oscillating dipole with electric dipole moment

$$\vec{p}_E(t) = -e\vec{x}(t) \quad (2.15)$$

which, in turn, radiates a light wave, called a secondary wave. If the electron executes a purely forced oscillation and no other degrees of freedom are involved, this wave has the same frequency as the exciting light wave. We then call the re-radiation elastic light scattering. Due to the phase delay between the exciting field and the charge oscillation (eq. (2.14)), the emitted light wave also has a phase delay with respect to the exciting wave.

Because a medium through which the light wave propagates is composed of many atoms, the light absorption in the individual atoms leads to macroscopically measurable consequences. Because a part of the light energy is transformed into heat during the propagation of the light wave through the medium, the intensity decays with increased propagation distance. If the medium is homogeneous, the relative decrease

$\frac{dI_p}{dz} \frac{1}{I_p(z)}$ along a small interval dz of the light path is independent of z :

$$\frac{dI_p}{dz} \frac{1}{I_p(z)} = -\alpha . \quad (2.16)$$

The constant α is called absorption coefficient. It is measured in m^{-1} or cm^{-1} . Integrating eq. (2.16) yields an exponential decay of intensity with increased propagation distance z in the medium:

$$I_p(z) = I_{p,0} e^{-\alpha z} \quad (\text{Beer's law}) . \quad (2.17)$$

In general, the absorption coefficient is frequency-dependent. It exhibits maxima at frequencies for which the absorption is resonant. Because the resonance frequencies are controlled by the electronic structure of the medium, the frequency dependence of α is specific for the material. For instance, while some materials, e.g., glass, are transparent ($\alpha \approx 0$) in the visible spectral range, most materials absorb ($\alpha > 0$) there. Similarly, the re-radiation of light by the individual atoms has consequences that are macroscopically measurable. At each point in space, the portion of the exciting light wave that has not been absorbed and the secondary waves are superimposed. Because of the phase delay φ between the exciting field and the charge displacement, the phase fronts created by this superposition also experience a delay with respect to the exciting wave. Because this delay takes place in each individual layer of the medium in the direction of propagation, the accumulated effect is that the phase velocity c in the medium is reduced with respect to the vacuum light velocity c_0 . The factor by which c is reduced is called the refractive index:

$$c = \frac{c_0}{n} \quad (\text{speed of light in a medium}) . \quad (2.18)$$

The refractive index changes the phase velocity, but not the angular frequency ω of the light wave. Therefore and due to eq. (2.4),

$$k = \frac{\omega}{c} = \frac{n\omega}{c_0} = nk_0 \quad (2.19)$$

and

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{nk_0} = \frac{\lambda_0}{n} . \quad (2.20)$$

Here, k_0 and λ_0 denote the values of the wave number and wavelength in vacuum, respectively. When light transits from vacuum into a medium, its velocity is, thus, reduced by a factor of n , the wave vector is stretched by n and the wavelength is shortened by the same factor. Similarly to the absorption coefficient, the refractive index is specific for the material (e.g., $n \approx 1.5$ for glass in the visible spectrum range) and, in general, frequency-dependent. The different refractive indices of media and vacuum show themselves macroscopically, for instance, in the phenomenon of light refraction during the transition of a light wave from the vacuum into the medium or in the opposite direction.

So far, we have considered light scattering and absorption separately from each other. Both effects can be formally summarized in complex notation by extending eq. (2.9) by a factor that describes the absorption-induced amplitude decay with propagation distance z (eq. (2.17)), and by replacing k from eq. (2.19) by nk_0 :

$$\tilde{\vec{E}}(z, t) = \vec{E}_0 e^{-\frac{\alpha}{2}z} e^{i(nk_0 \cdot z - \omega t)} . \quad (2.21)$$

This representation becomes more concise when we define a complex refractive index $\tilde{n}$ through

$$\tilde{n} := n + i\kappa \quad (2.22)$$

with the extinction coefficient

$$\kappa := \frac{\alpha}{2k_0} . \quad (2.23)$$

From eq. (2.21) and using eqs. (2.22) and (2.23), we obtain the more compact expression

$$\tilde{\vec{E}}(z, t) = \vec{E}_0 e^{i(\tilde{n}k_0 \cdot z - \omega t)} . \quad (2.24)$$

When, in analogy to eq. (2.19), a complex wavenumber $\tilde{k}$ is defined by

$$\tilde{k} = \tilde{n} \frac{\omega}{c_0} = \tilde{n}k_0 , \quad (2.25)$$

the representation becomes still shorter:

$$\tilde{\vec{E}}(z, t) = \vec{E}_0 e^{i(\tilde{k} \cdot z - \omega t)} . \quad (2.26)$$

Here, the absorption-induced decay of the amplitude is described by the imaginary part of the complex wavenumber $\tilde{k}$.

2.2.2.2 Microscopic and macroscopic optical responses

In the previous subsection, we have argued that the electron oscillation in the atoms, which is a microscopic response to the optical field, leads to a retardation and absorption of the light wave in the medium, i.e., macroscopically measurable effects. In the present subsection, we will study the connection between the microscopic and macroscopic optical responses of the medium in more detail. To this end, we will start from the equation of motion (eq. (2.11)) of the Lorentz oscillator. We slightly rewrite eq. (2.12) to obtain compact expressions later on:

$$\tilde{\vec{x}}(t) = \tilde{\vec{x}}_0 e^{i(\tilde{k} \cdot \vec{r} - \omega t)} \quad (2.27)$$

with the complex-valued amplitude $\tilde{\vec{x}}_0 = \vec{x}_0 e^{i\varphi}$. The phase factor $e^{i\varphi}$ is now included in the amplitude and $\vec{r}$ is again the position of the atom. When we enter the ansatz from eq. (2.27) into the equation of motion (2.11), we obtain the complex-valued amplitude

$$\tilde{\vec{x}}_0 = \frac{-e\vec{E}_0}{m_e} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} . \quad (2.28)$$

The displacement $\tilde{x}(t)$ of the electron is then obtained as the real part of the complex-valued function $\tilde{\tilde{x}}(t)$ calculated using eqs. (2.27) and (2.28). Like eq. (2.13), eq. (2.28) shows that the oscillation amplitude is especially large for $\omega \approx \omega_0$, i.e., near the resonance frequency, and small away from it. The displacements $\tilde{x}(t)$ of the negatively charged electron relative to the position of the atom core give rise to an oscillating dipole moment $\tilde{p}_{E,\text{res}}(t) = -e\tilde{x}(t)$. This can also be written in complex notation as

$$\tilde{p}_{E,\text{res}}(t) = -e\tilde{\tilde{x}}(t) . \quad (2.29)$$

The index “res” is supposed to remind us that the amplitude of this dipole moment becomes, according to eq. (2.28), maximal at the resonance frequency. In what follows, n_A denotes the number of atoms per unit volume. The contribution made to the dielectric polarization $\tilde{P}$ by the dipole moments $\tilde{p}_{E,\text{res}}$ generated by the oscillation is:

$$\tilde{P}_{\text{res}} = n_A \tilde{p}_{E,\text{res}} = -n_A e \tilde{\tilde{x}} . \quad (2.30)$$

We assume that all atoms oscillate in phase, i.e., we consider a volume smaller than λ of the light wave. Additionally, we subsume into a term $\tilde{P}_{\text{nonres}}$ the contributions from charge oscillations that are nonresonant at $\omega \approx \omega_0$, e.g., of ions of the crystal lattice. The total dielectric polarization can thus be written as:

$$\tilde{P} = \tilde{P}_{\text{nonres}} + \tilde{P}_{\text{res}} . \quad (2.31)$$

If we express the polarizations by the associated susceptibilities χ and the electric field according to

$$\tilde{P} = \varepsilon_0 \chi \tilde{\tilde{E}} , \quad (2.32)$$

we obtain

$$\chi = \chi_{\text{nonres}} + \chi_{\text{res}} , \quad (2.33)$$

where χ_{res} , because of eqs. (2.28)–(2.33), has the frequency dependence

$$\chi_{\text{res}}(\omega) = \frac{n_A e^2}{\varepsilon_0 m_e} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} . \quad (2.34)$$

As known from electrodynamics, the dielectric polarization is connected to the electric displacement $\tilde{D}$ through the equation

$$\tilde{D} = \varepsilon_0 \tilde{\tilde{E}} + \tilde{P} . \quad (2.35)$$

Inserting eq. (2.32) gives

$$\tilde{D} = \varepsilon_0 \varepsilon \tilde{\tilde{E}} , \quad (2.36)$$

where

$$\varepsilon = 1 + \chi \quad (2.37)$$

is the relative permittivity, also called the dielectric constant of the medium. From eq. (2.36) and the fact that in the absence of free charges a $\vec{D}$ field perpendicular to the surface of the medium has the same value inside and outside the medium, we obtain

$$\vec{E} = \frac{\vec{E}_{\text{ext}}}{\varepsilon}, \quad (2.38)$$

where $\vec{E}_{\text{ext}}$ is the electric field outside the medium. The term $1/\varepsilon$ thus gives the factor by which the electric field in the medium differs from that outside, if the fields are directed perpendicular to the surface of the medium. This difference is a consequence of the electric counter field that is generated by the charge displacements in the medium, which is superimposed on the external electric field. Like χ , ε is also frequency-dependent. For instance, for the Lorentz oscillators described by eq. (2.34), it has the form:

$$\varepsilon(\omega) = 1 + \chi_{\text{nonres}} + \frac{n_A e^2}{\varepsilon_0 m_e} \frac{1}{(\omega_0^2 - \omega^2 - i\gamma\omega)}. \quad (2.39)$$

To stress the frequency dependence of ε , $\varepsilon(\omega)$ is often called the dielectric function. The frequency dependence described in eq. (2.39) is plotted in Fig. 2.3 for the example of the D_2 line of sodium. As expected, it shows a resonance at the eigenfrequency ω_0 , with a width determined by the damping rate γ .

Real solids have more than one optical resonance as they support vibrational and electronic excitations at various frequencies. For many solids, their dielectric functions can be looked up in tables. Alternatively, they can be modelled as superpositions

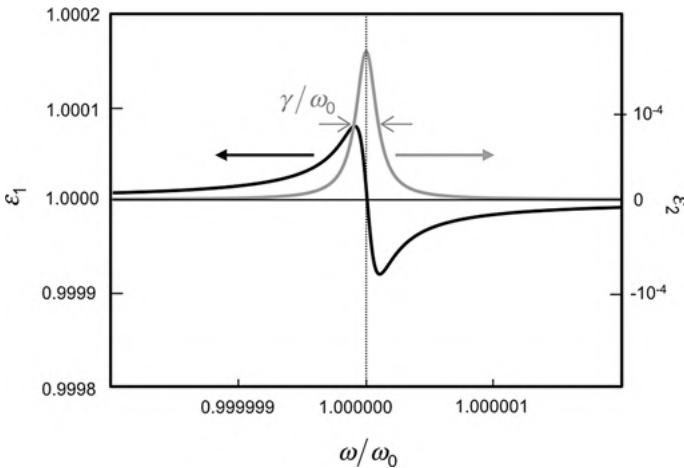


Fig. 2.3: Frequency dependence of the dielectric function $\varepsilon(\omega)$ of a medium with Lorentz oscillators with density $n_A = 10^{17} \text{ m}^{-3}$, eigenfrequency $\omega_0 = 3.20 \times 10^{15} \text{ s}^{-1}$, damping rate $\gamma = 6.28 \times 10^8 \text{ s}^{-1}$, and $\chi_{\text{nonres}} = 0$ (eq. (2.39)). These numbers are valid for the strongest hyperfine component of the D_2 line of a gas of sodium atoms [15]. ε_1 and ε_2 are the real and imaginary parts of the dielectric function $\varepsilon(\omega)$, respectively.

of Lorentz oscillator terms, one for each optical excitation in the solid, and each with their own resonance frequency ω_{0j} , damping rate γ_j and spectral weight f_j :

$$\varepsilon(\omega) = 1 + \frac{n_A e^2}{\varepsilon_0 m_e} \sum_j \frac{f_j}{(\omega_{0j}^2 - \omega^2 - i\gamma_j \omega)}. \quad (2.40)$$

A comparison with eq. (2.39) reveals the origin of the term χ_{nonres} . It stands for those Lorentz oscillator terms in eq. (2.40) that are off-resonance at a given wavelength ω .

Once the dielectric function has been determined from tables or through a model calculation, the refractive index n and the absorption coefficient α of the respective materials can be calculated. This useful connection can be derived from Maxwell's equations

$$\nabla \cdot \vec{D} = \rho_c \quad (2.41)$$

$$\nabla \cdot \vec{B} = 0 \quad (2.42)$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (2.43)$$

$$\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}, \quad (2.44)$$

where $\vec{H}$ is the magnetic field, ρ_c the density of free charges, and $\vec{j}$ the electric current density. Let us assume that our medium is a dielectric, i.e., no currents can flow in it (current density $\vec{j} = \vec{0}$). Furthermore, the medium is assumed to be nonmagnetic with permeability $\mu_M = 1$ and, hence, we have $\vec{B} = \mu_0 \vec{H}$, where μ_0 is the vacuum permeability. By combining eqs. (2.43) and (2.44), we obtain:

$$\nabla \times (\nabla \times \vec{E}) = -\mu_0 \varepsilon_0 \varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}. \quad (2.45)$$

If we take into account the vector identity $\nabla \times (\nabla \times E) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$ and assume no free charges ($\rho_c = 0$), we obtain the equation

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}. \quad (2.46)$$

Entering a plane harmonic wave of the form in eq. (2.26) into eq. (2.46) and exploiting $c_0 = 1/\sqrt{\mu_0 \varepsilon_0}$ gives:

$$\tilde{k}^2 = \varepsilon \frac{\omega^2}{c_0^2}. \quad (2.47)$$

A comparison with eq. (2.25) gives for the refractive index

$$\varepsilon = \tilde{n}^2. \quad (2.48)$$

Equation (2.48) makes an interesting physical statement: The change of the electric field caused by the medium as described by ε is directly linked to the phase velocity

and the absorption of the light wave in this medium, as described by $\tilde{n} = n + i\kappa$. From eq. (2.48), the following relations are derived between the real and imaginary parts of the two quantities $\varepsilon = \varepsilon_1 + i\varepsilon_2$ and $\tilde{n} = n + i\kappa$ (eq. (2.22)):

$$\varepsilon_1 = n^2 - \kappa^2 \quad (2.49)$$

and

$$\varepsilon_2 = 2n\kappa. \quad (2.50)$$

The conversion formulas in the opposite direction are

$$n = \frac{1}{\sqrt{2}} \sqrt{\varepsilon_1 + \sqrt{\varepsilon_1^2 + \varepsilon_2^2}} \quad (2.51)$$

and

$$\kappa = \frac{1}{\sqrt{2}} \sqrt{-\varepsilon_1 + \sqrt{\varepsilon_1^2 + \varepsilon_2^2}}. \quad (2.52)$$

Using these two equations and eq. (2.23), the refractive index n and the absorption coefficient α can be calculated from the dielectric function $\varepsilon(\omega)$. An example is shown in Fig. 2.4, in which n and κ have been calculated from the dielectric function shown in Fig. 2.3.

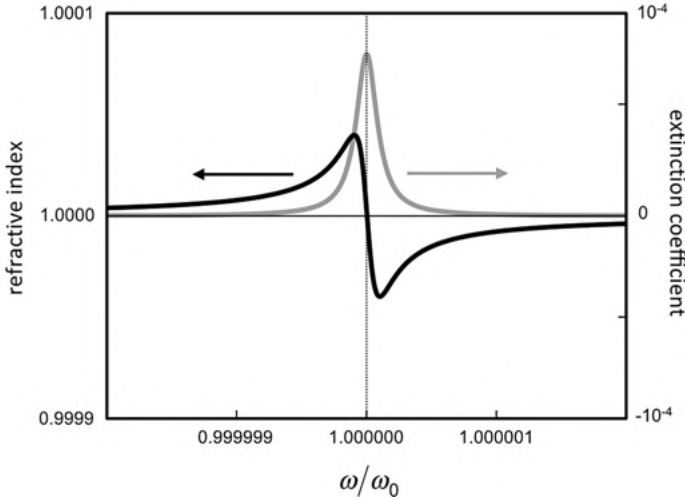


Fig. 2.4: Real refractive index n and extinction coefficient κ calculated from the dielectric function according to eqs. (2.51) and (2.52) for the same medium of Lorentz oscillators as in Fig. 2.3

2.2.2.3 Light reflection

So far, we have dealt with light propagation and absorption in homogeneous media. We know from electrodynamics that light incident on the interface between two media

1 and 2 is partially reflected by the interface. For instance, a fraction of approximately 4% of the light power is reflected for perpendicular incidence of a light wave from air onto a glass surface. In what follows, we will deal with the question how one can calculate the reflected fraction. The ratio of the time-averaged reflected light power $\bar{P}_r$ to the time-averaged total incident power $\bar{P}_i$

$$R_y := \frac{\bar{P}_r}{\bar{P}_i} \quad (2.53)$$

is called reflectivity. The corresponding fraction transmitted through the interface

$$T_y := \frac{\bar{P}_t}{\bar{P}_i} \quad (2.54)$$

is called transmissivity. If the participating media are transparent and thus no absorption takes place,

$$R_y + T_y = 1 \quad (2.55)$$

must hold due to energy conservation. Because of eqs. (2.6) and (2.8), the reflectivity is connected to the electric field amplitudes of the incident and reflected wave by

$$R_y = \left(\frac{E_{0,r}}{E_{0,i}} \right)^2. \quad (2.56)$$

The amplitude ratio appearing in eq. (2.56) can also be derived from Maxwell's equations for arbitrary angles of incidence β of the light wave. For perpendicular incidence ($\beta = 0$) from air onto the surface of a medium with refractive index $\tilde{n} = n + i\kappa$, one obtains:

$$R_y(\beta = 0) = \left| \frac{\tilde{n} - 1}{\tilde{n} + 1} \right|^2 = \left| \frac{n + i\kappa - 1}{n + i\kappa + 1} \right|^2 = \frac{(n - 1)^2 + \kappa^2}{(n + 1)^2 + \kappa^2}. \quad (2.57)$$

Equation (2.57) shows that $R_y \approx 1$ for $\kappa \gg n$. Therefore, strongly absorbing media also have a high reflectivity. Furthermore, if the absorption coefficient α and thus the extinction coefficient κ are strongly dependent on the wavelength, this also applies to the reflectivity.

2.2.2.4 Classical light emission

In Subsection 2.2.2.1, it was mentioned that an excited atom can re-radiate the energy taken up from the light wave. In this way, it behaves like every oscillating dipole which radiates electromagnetic waves due to its charge oscillation. The oscillation energy is thus gradually transformed into electromagnetic field energy, which is transported away from the atom. The loss of oscillation energy limits the time during which an atom can oscillate after the passage of the exciting electromagnetic wave train. In what follows, we will calculate the decay time of the oscillation. The mechanical oscillation energy of the Lorentz oscillator (Section 2.2.2.1) is given by:

$$E_{\text{osc}} = \frac{1}{2} m_e \omega_0^2 x_0^2. \quad (2.58)$$

As known from electrodynamics, the power that is radiated by an oscillating dipole with charges $\pm e$, averaged over one oscillation period, is given by

$$\bar{P}_{\text{em}} = \frac{e^2 \omega_0^4 x_0^2}{12\pi\epsilon_0 c^3}. \quad (2.59)$$

If the lost energy is not replaced from the outside, the oscillation energy of the dipole decreases at the rate described by eq. (2.59). According to eq. (2.58), this means that the oscillation amplitude x_0 decays. The relative reduction of the oscillation energy is

$$\frac{dE_{\text{osc}}}{dt} \frac{1}{E_{\text{osc}}} = \frac{\bar{P}_{\text{em}}}{E_{\text{osc}}} = -\frac{e^2 \omega_0^2}{6\pi\epsilon_0 m_e c^3}. \quad (2.60)$$

Solving the differential eq. (2.60) gives an exponential decay of the oscillation energy:

$$E_{\text{osc}}(t) = E_{\text{osc}}(t=0) \cdot e^{-\frac{t}{\tau_{\text{R}}^{(\text{cl})}}}. \quad (2.61)$$

Here, $\tau_{\text{R}}^{(\text{cl})} = \frac{e^2 \omega_0^2}{6\pi\epsilon_0 m_e c^3}$ is called the radiative lifetime of the classical dipolar oscillation.

As mentioned at the end of Section 2.2.1, the classical description neglects the quantization of the light field into photons, which becomes relevant at lower light intensity and if one considers the energy exchange between quasiparticles, such as crystal electrons, and the light field. The latter is the subject of the following section.

2.2.3 Quantum physics of optical transitions

Our classical treatment of the microscopic optical properties of solids in the previous section has enabled us to achieve a simple understanding of important phenomena such as light absorption. However, there are optical phenomena in solids which cannot be described in this way and which can only be explained using quantum physics. One example is stimulated emission as employed by lasers. In the present section, we will discuss the quantum physics of optical transitions using a simple model system, the two-level system. For simplicity, we will initially do this in a semi-classical approximation, in which matter is treated quantum-mechanically and the light field classically. Subsequently, we will describe spontaneous emission in a simplified quantum-optical picture.

2.2.3.1 The two-level system

As mentioned in Section 2.2.2.1, an optically driven electron oscillation in an atom represents, from a quantum physical perspective, an oscillating spatial displacement of the electron wave function relative to the positively charged atom core. In what follows, we will show that this displacement can be described by an appropriate superposition of the energy eigenfunctions of the electron. We keep this treatment as simple as possible by assuming that only two states of the electron in the atom contribute to

this superposition and that no damping processes are involved. The two-level system is composed of a state ψ_1 with energy E_1 and a second state ψ_2 with energy $E_2 > E_1$ (Fig. 2.5). In the absence of light, ψ_1 and ψ_2 are energy eigenstates of the system and thus obey the time-independent Schrödinger equation:¹

$$\hat{H}_0 \psi_j = E_j \psi_j, \quad (2.62)$$

where $\hat{H}_0$ is the Hamilton operator of the unperturbed two-level system without incident light. ψ_1 and ψ_2 have the time-dependent form

$$\tilde{\psi}_j(\vec{r}, t) = \psi_j(\vec{r}) \exp\left(-\frac{iE_j t}{\hbar}\right), \quad j = 1, 2. \quad (2.63)$$

When ψ_1 and ψ_2 are coupled by the alternating electric field of a light wave at frequency ω , they cease to be true eigenstates. The time evolution of the system is then described by the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \tilde{\psi}}{\partial t} = (\hat{H}_0 + \hat{H}') \tilde{\psi}, \quad (2.64)$$

where $\tilde{\psi}$ is a time- and position-dependent wave function. $\hat{H}'$ from eq. (2.64) is the operator of the perturbation by light. In classical electrodynamics, the electric field $\vec{E}$ causes an energy shift of a dipole $\vec{p}_E$ by $-\vec{p}_E \cdot \vec{E}$. The perturbation operator $\hat{H}'$ can be expressed in terms of this energy shift by replacing the two involved classical quantities, $\vec{p}_E$ and $\vec{E}$, with their associated operators. In what follows, we will use the semiclassical approximation, in which particles are treated quantum mechanically while the field is that of a classical electromagnetic wave. The perturbation operator is then:

$$\hat{H}' = -\vec{E} \cdot \hat{\vec{p}}_E, \quad (2.65)$$

where

$$\hat{\vec{p}}_E = -e\hat{\vec{x}} \quad (2.66)$$

is the electric dipole operator. In agreement with eq. (2.2), we write the field at the position $\vec{r} = \vec{0}$ in an atom as

$$\vec{E}(t) = \frac{1}{2} \vec{E}_0 (e^{i\omega t} + e^{-i\omega t}). \quad (2.67)$$

$\tilde{\psi}$ can be written as a superposition of the two states ψ_1 and ψ_2 of the unperturbed two-level system:

$$\tilde{\psi} = c_1 \psi_1 + c_2 \psi_2 \quad (2.68)$$

with the time-dependent coefficients $c_1(t)$ and $c_2(t)$. Entering the ansatz of eq. (2.68) into the Schrödinger eq. (2.64), multiplying with $\tilde{\psi}^*$ from the left side, integrating over

¹ We use a hat above a symbol (such as in $\hat{A}$) to indicate quantum mechanical operators.

the volume of the system and taking into account eq. (2.62) yields a system of two coupled differential equations:

$$i\hbar\dot{c}_1 = (E_1 + \widehat{H}'_{11})c_1 + \widehat{H}'_{12}c_2 \quad \text{and} \quad i\hbar\dot{c}_2 = \widehat{H}'_{21}c_1 + (E_2 + \widehat{H}'_{22})c_2 \quad (2.69)$$

with the matrix elements

$$\widehat{H}'_{jk} = \vec{E} \cdot \vec{\mu}_{jk}, \quad (2.70)$$

where

$$\vec{\mu}_{jk} = \int \psi_j^* e\vec{x}\psi_k d^3\vec{x} \quad (2.71)$$

is the dipole matrix element between the states ψ_j and ψ_k for $j, k = 1, 2$ (also called the transition dipole moment). In many cases, the diagonal elements $\vec{\mu}_{11}$ and $\vec{\mu}_{22}$ vanish for symmetry reasons, as we assume in the following. Of course, we have to assume additionally that the selection rules allow the transitions $\psi_1 \rightarrow \psi_2$ and $\psi_2 \rightarrow \psi_1$ to occur, which implies, in particular, $\vec{\mu}_{12} \neq \vec{0}$ and $\vec{\mu}_{21} \neq \vec{0}$. Using, moreover, the transformation $c_j =: d_j \exp\left(\frac{-iE_j t}{\hbar}\right)$, eqs. (2.69) read

$$i\hbar\dot{d}_1 = \widehat{H}'_{12}d_2 e^{-i\frac{(E_2-E_1)t}{\hbar}} \quad \text{and} \quad i\hbar\dot{d}_2 = \widehat{H}'_{21}d_1 e^{-i\frac{(E_1-E_2)t}{\hbar}}. \quad (2.72)$$

By entering the eqs. (2.70) and (2.67) into eq. (2.72), restricting ourselves to resonant excitation $\hbar\omega = E_2 - E_1$, neglecting rapidly rotating terms at frequency 2ω (rotating wave approximation) and transforming back to the coefficients c_1 and c_2 , one obtains the following solutions:

$$c_1 = e^{-\frac{iE_1 t}{\hbar}} \cos(\omega_{\text{Rabi}} t) \quad (2.73)$$

and

$$c_2 = e^{-\frac{iE_2 t}{\hbar}} \sin(\omega_{\text{Rabi}} t), \quad (2.74)$$

where $\omega_{\text{Rabi}} = \frac{\vec{E}_0 \cdot \vec{\mu}_{12}}{\hbar}$ is called the Rabi frequency. Thus we have determined the time dependence of the electron state $\tilde{\psi}$ from eq. (2.68). The probability density oscillates between state ψ_1 and state ψ_2 . Particularly instructive is the expectation value $\langle \vec{x} \rangle$ of the electron position, which can be calculated using eqs. (2.68), (2.71), (2.73) and (2.74):

$$\langle \vec{x} \rangle = \int \tilde{\psi}^* \vec{x} \tilde{\psi} d^3\vec{x} = -\frac{\vec{\mu}_{12}}{e} \sin(\omega_{\text{Rabi}} t) \sin\left(\frac{(E_2 - E_1)t}{\hbar}\right). \quad (2.75)$$

This equation tells us that the superposition of the two wave functions ψ_1 and ψ_2 leads to an oscillation of the average electron position back and forth in space at the frequency $\frac{(E_2-E_1)}{\hbar}$. This corresponds to the frequency ω_0 at which the Lorentz oscillator of eq. (2.12) oscillates back and forth. In addition, the amplitude of the quantum mechanical oscillation is modulated at the Rabi frequency ω_{Rabi} , which is typically much smaller than $\frac{E_2-E_1}{\hbar}$ and tunable by the electric field strength $\vec{E}_0$ of the light wave. The physical meaning of this frequency becomes clear when we calculate the probabilities $|c_j|^2$ of finding the electron in the lower or upper state via eqs. (2.73) and (2.74):

$$|c_1|^2 = \cos^2(\omega_{\text{Rabi}} t) \quad (2.76)$$

and

$$|c_2|^2 = \sin^2(\omega_{\text{Rabi}} t) . \quad (2.77)$$

In the absence of damping processes, the electron goes back and forth between the lower and the upper state at the frequency $2 \cdot \omega_{\text{Rabi}}$ – a process called Rabi oscillation. This sequence of light-induced excitation and de-excitation between the two states is a purely quantum-physical effect and is caused by the light-induced coupling of the two states. It enables one to deliberately transfer the ground state ψ_1 into a superposition of both states ψ_1 and ψ_2 via choosing the time t that the light wave acts on the two-level system (see also Section 3.4.3). It is also responsible for the oscillation of the average charge position in eq. (2.75).

The oscillation vanishes at those moments at which the electron is found with certainty in one of the two states. In contrast, it is maximal when $|c_1(t)|^2 = |c_2(t)|^2 = \frac{1}{2}$. This behavior is distinct from the behavior of the classical Lorentz oscillator, whose oscillation amplitude would, in the absence of damping, increase monotonically with time, taking up energy from the light field in every oscillation cycle of the field. The light-induced transitions between the states in the Rabi oscillation act back on the light field due to energy conservation. During the transition $\psi_1 \rightarrow \psi_2$, energy is absorbed from the light field, thus reducing the light power (Fig. 2.5). In the reverse transition $\psi_2 \rightarrow \psi_1$, energy is restored to the light field, a phenomenon that is called stimulated emission as it does not occur spontaneously but is induced by the light field.

Normally, damping processes prevent an observation of complete cycles of a Rabi oscillation. In this case, the damping rate is much larger than the Rabi frequency and the damping-induced de-excitation of the two-level system dominates. It can be shown that for stationary excitation conditions, the occupation of the upper state then becomes

$$|c_2|^2 \approx \left(\frac{\omega_{\text{Rabi}}}{2\gamma} \right)^2 , \quad (2.78)$$

i.e., $|c_2|^2 \ll 1 \approx |c_1|^2$. The system always tends to this equilibrium value, if it has either been in the lower or in the upper level at the beginning of the light illumination. In addition, it follows from eq. (2.78) that in stationary excitation conditions $|c_2|^2$ is approximately proportional to ω_{Rabi}^2 and thus to $|\vec{E}_0|^2$, which is also true for the oscillation energy of the classical Lorentz oscillator as described in eq (2.58). Hence, strong damping restores the classical behavior of electrons in an atom.

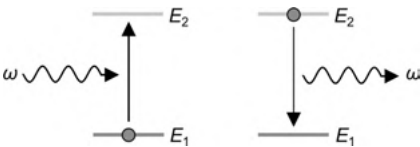


Fig. 2.5: Optical transitions of an electron between two energy levels in an atom due to absorption (left) and emission (right) of light.

2.2.3.2 Spontaneous and stimulated emission

As known from quantum physics, light can be decomposed into tiny portions called light quanta or photons. They have the photon energy

$$E_\gamma = \hbar\omega \quad (2.79)$$

and photon momentum

$$\vec{p}_\gamma = \hbar\vec{k}_\gamma . \quad (2.80)$$

The research field that deals with the properties of photons and their interaction with matter is called quantum optics. Here, the photons are the excitations of the light field. In the quantum optical treatment of light absorption and emission, it is assumed that these two processes involve transitions of atoms between two quantum states. An atom absorbs a photon by jumping into a higher energy level with energy E_2 (Fig. 2.5). The emission of a photon involves a jump into a lower level with energy E_1 . If we neglect – in good approximation – the effects of photon absorption or emission on the center of mass motion of the atom, we can identify the energies E_1 and E_2 with those of the jumping electron in the atom. Energy conservation requires that the energy of the photon equals the energy difference between the initial and the final level for both directions of a jump:

$$\hbar\omega = E_2 - E_1 . \quad (2.81)$$

In order to determine the temporal rates at which electron jumps occur, we choose a simple description using Einstein coefficients. This description can be derived from a more demanding quantum theoretical treatment. We express the number of absorption events by the rate at which the occupation N_1 of the lower level changes due to upward transitions:

$$\frac{dN_1}{dt} = -B_{12}N_1u(\omega) . \quad (2.82)$$

Here B_{12} is the Einstein B coefficient which expresses how sensitively an electron in the lower level reacts to a perturbation by a photon. The function $u(\omega)$ is the spectral energy density of light at frequency ω . It is proportional to the intensity of light or – in quantum optical language – to the number density of photons at frequency ω per volume. Here, we consider only the case of resonant absorption, meaning that ω is determined by the resonance condition (2.81). In an analogous fashion, we can express the rate at which the occupation N_2 of the upper level changes due to stimulated emission:

$$\frac{dN_2}{dt} = -B_{21}N_2u(\omega) . \quad (2.83)$$

Here B_{21} is the Einstein B coefficient for stimulated emission, which expresses how sensitively an electron in the upper level reacts to a perturbation by a photon. It is possible to show that the two Einstein B coefficients are closely linked, i.e., the ratio B_{21}/B_{12} is independent of the material. In stimulated emission, light that induces

the downward transition adds another photon and thus increases the number of photons. It can be shown that the light added by stimulated emission always has the same optical phase as the incident light, which means that it oscillates coherently with the latter. In Section 2.2.2.4, we have seen that a classical oscillating dipole spontaneously loses energy through electromagnetic radiation. This phenomenon also exists in quantum optics and is known here as spontaneous emission. It can be understood as stimulated emission caused by vacuum fluctuations of the electromagnetic field. In the Einstein description, the rate at which the occupation N_2 of the upper level changes due to spontaneous emission has the form

$$\frac{dN_2}{dt} = -A_{21}N_2. \quad (2.84)$$

Here, A_{21} is the Einstein A coefficient for spontaneous emission, which expresses how sensitively an electron in the upper level reacts to the perturbation by the vacuum fluctuations. As the vacuum fluctuations are not included in the spectral energy density, $u(\omega)$ is absent in eq. (2.84). The solution of the equation shows that spontaneous emission leads to an exponential decay of the occupation of the upper level:

$$N_2(t) = N_2(0)e^{-A_{21}t} = N_2(0)e^{-\frac{t}{\tau_R}}, \quad (2.85)$$

where $\tau_R = A_{21}^{-1}$ is called the radiative lifetime of the transition. One thus obtains an exponential decay of the same form as in the classical decay of the oscillation energy (eq. (2.61)). It can be shown that $A_{21}(\omega)/B_{21}(\omega)$ is also independent of the material, i.e., the Einstein coefficient for spontaneous emission is related to that for stimulated emission.

An important application of stimulated emission are lasers. The word LASER is an acronym for “light amplification by stimulated emission of radiation.” Lasers are based on an excitation of electrons (called pumping) within a medium, so that the occupation N_2 becomes larger than the occupation N_1 . This is called inversion. Provided that the values of B_{21} and B_{12} are not too different, the consequence of the inversion is, according to eqs. (2.82) and (2.83), that the stimulated emission rate exceeds the absorption rate, resulting in an amplification of the light field created by the first spontaneous emission processes rather than a light attenuation as in ordinary absorbing media. In this case, the medium consisting of excited atoms is called a gain medium. Moreover, for lasers, a resonator is set up by placing the gain medium between two mirrors (Fig. 2.6). One of the mirrors, the high reflector, is in general highly reflective, and the other one, the output coupler, is not as highly reflective, but slightly transparent for the photons. Because the photons circulate between the mirrors several times before leaving the resonator through the output coupler, the spectral energy density in the gain medium is high and, hence, the stimulated emission is much larger than the spontaneous one. If the stimulated emission dominates, the light waves emitted at various positions in the gain medium are coherent resulting in laser emission.

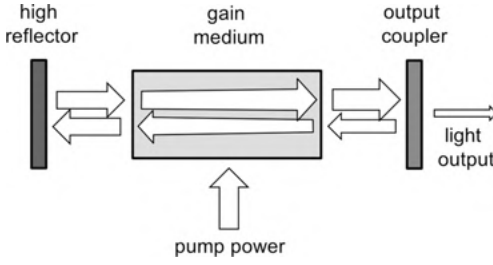


Fig. 2.6: Laser resonator. Light circulates in the resonator by repeatedly passing through a gain medium and being reflected by mirrors. The gain medium is optically or electrically excited by an external energy source (a process called pumping) and amplifies the light during each of its passages. One of the mirrors has a high reflectivity (high reflector). A fraction of the light leaves the resonator through the other mirror with the lower reflectivity (output coupler).

In a fashion analogous to the light attenuation in an absorbing medium, which is expressed by the absorption coefficient α (eq. 2.16), we can express the optical amplification at the frequency ω using the gain coefficient g_ω :

$$dI_p = +g_\omega I_p(z) dz . \quad (2.86)$$

The positive sign in the differential reflects that the intensity grows with distance. Integration of the equation gives:

$$I_p(z) = I_{p,0} e^{g_\omega z} . \quad (2.87)$$

A condition for stable laser emission is that the light intensity in the resonator does not change over time. This is the case if the intensity increase caused by the amplification exactly compensates the light losses due to the non-perfect reflectivities of the mirrors and other losses. For a complete round-trip in the resonator through the gain medium of length L with mirrors having reflectivities $R_{y,1}$ and $R_{y,2}$, this condition reads

$$e^{(g_\omega - \alpha_{\text{loss}}) \cdot 2L} \cdot R_{y,1} \cdot R_{y,2} = 1 . \quad (2.88)$$

The attenuation coefficient α_{loss} takes into account scattering losses as well as absorption by impurities in the medium. The condition in eq. (2.88) can be written in the form

$$g_{\text{th}} := g_\omega = \alpha_{\text{loss}} - \frac{1}{2L} \ln(R_{y,1} R_{y,2}) . \quad (2.89)$$

This equation defines the threshold gain g_{th} , which is the value where laser emission sets in.

2.3 Interband transitions in solids

In Subsections 2.2.2.1 and 2.2.3.1, we have dealt with optical transitions between the electron states in atoms. These states are spatially confined to the individual atoms

and energetically discrete. In contrast, the electron states in crystalline solids are spatially extended and arranged in quasi-continuous energy continua called energy bands. If the initial and final state of an optical transition lie in different bands, the transition is called an interband transition. The present section deals with interband transitions as they appear in absorption and emission spectra of solids.²

2.3.1 Interband absorption

Absorption of light by interband transitions of electrons in solids is called interband absorption. In the present section, we will learn that the properties of interband absorption are controlled by properties of the initial and final states of the transition. Measurements of interband absorption thus reveal important information about electronic properties such as excited-state energies and the densities of states of those bands.

2.3.1.1 Interband transitions of single electrons

Similarly to the case of single atoms, the alternating electric field of an incident light wave can excite an electron oscillation in solids. Let us consider a single electron occupying a state in a solid. In many cases, the light field couples this initial state predominantly to one state of higher energy. As in Section 2.2.3.1, the resulting oscillation is described quantum physically by a superposition of the two states. Again, the field-induced superposition is accompanied by a transition of the electron from the occupied initial state to the higher empty state. In crystalline solids, these states are extended Bloch waves:

$$\psi_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{V}} u_{\vec{k}}(\vec{r}) e^{i\vec{k} \cdot \vec{r}}. \quad (2.90)$$

Here, V is the volume of the solid, $u_{\vec{k}}(\vec{r})$ is a function with the same periodicity as the lattice and $\vec{r}$ is the position of the electron. We denote the initial state as $\psi_{\vec{k}_i}$ and the final state as $\psi_{\vec{k}_f}$. The energy required for the excitation is absorbed from the light field as a photon with energy $\hbar\omega$. For instance, Fig. 2.7 shows a transition that promotes an electron from an initial state with an energy E_i in a fully occupied lower band into a final state with the energy E_f in an initially empty upper band. Since both states are Bloch states, they are both spatially extended.

The possible combinations of initial and final states are not arbitrary. As in the atomic case, this has several reasons. First, the initial state must be occupied and the final state must be empty for an optical transition to occur. The law of energy conservation and the selection rules must be fulfilled. Due to the lattice periodicity of the Bloch states, also the law of momentum conservation must be fulfilled. This law reads

² In this section, we will follow the excellent presentation of Ref. [15].

for the absorption of a photon:³

$$\hbar\vec{k}_f = \hbar\vec{k}_i + \hbar\vec{k}_\gamma . \quad (2.91)$$

Here, $\hbar\vec{k}_\gamma$ is the momentum of the exciting photon (eq. (2.80)). Equation (2.91) holds if the electron is not scattered by a phonon during the photon absorption. An optical transition for which eq. (2.91) holds is called a direct transition, and the corresponding absorption process is known as direct absorption. Because the wavelength of light is much larger than the lattice constant of the crystal, the photon momentum can, in general, be neglected compared to typical electron momenta. We therefore get, to a good approximation,

$$\hbar\vec{k}_f = \hbar\vec{k}_i , \quad (2.92)$$

i.e., direct transitions are almost vertical in the $E(\vec{k})$ diagram. For direct absorption, the law of energy conservation reads

$$E_f = E_i + \hbar\omega , \quad (2.93)$$

i.e., the energy required for the excitation originates completely from the absorption of the photon. These conditions for direct absorption (appropriate state occupation, selection rules, momentum and energy conservation) usually forbid direct transitions of single electrons within the same band at frequencies in the visible and near-infrared spectral ranges. In contrast, an excitation of the electron from an initial state in a lower band into a final state in a higher band is often allowed. An example of such an interband absorption has been shown schematically in real space in Fig. 2.7. The simultaneous momentum and energy conservation can best be illustrated in an $E(\vec{k})$ diagram of a solid such as Fig. 2.8. As in Fig. 2.7, the shading of the bands indicates that the lower band is occupied by electrons and the upper band is unoccupied. The arrow depicts

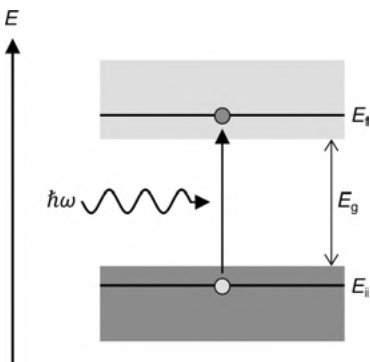


Fig. 2.7: Schematic diagram of an interband absorption process in a crystalline solid in real space. Absorption of a photon with energy $\hbar\omega$ promotes an electron from an initial state with energy E_i in a fully occupied lower band (shaded dark grey) to a final state with energy E_f in an initially empty upper band (shaded light grey). E_g is the band gap between the two bands.

³ We ignore possible contributions from reciprocal lattice vectors $\vec{G}_i$, which could add momenta $\pm\hbar\vec{G}_i$, but are mostly irrelevant.

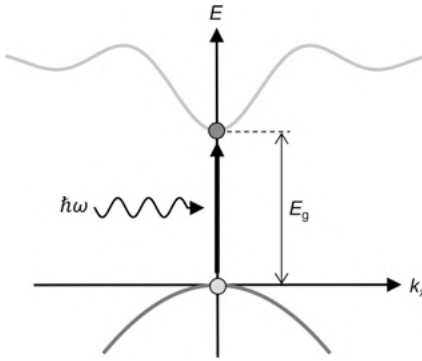


Fig. 2.8: An interband absorption process (arrow) displayed in an $E(\vec{k})$ diagram of a crystalline solid. An electron is optically excited from an initial state in the occupied lower (valence) band (dark grey line) to a final state in the unoccupied upper (conduction) band (light grey line).

the momentum and energy conservation for the special case of monochromatic excitation at the photon energy $\hbar\omega = E_g$. Moreover, we assume that the symmetry properties of the involved band states satisfy the selection rules for optical transitions.

Figure 2.8 depicts a situation in which the maximum of the lower occupied band is at the same point in $\vec{k}$ space as the minimum of the higher empty band. This means that the lowest photon energy at which direct transitions can occur is the band gap energy, $\hbar\omega = E_g$. Such a situation is found in direct semiconductors, in which the global maximum of the highest valence band and the global minimum of the lowest conduction band both lie at the same point in $\vec{k}$ space, usually at $\vec{k} = \vec{0}$. An example is the semiconductor gallium arsenide shown in Fig. 2.9. Direct transitions from states of the valence band maximum into empty states of the conduction band minimum can be excited at the Γ point of the Brillouin zone, i.e., at $\vec{k} = \vec{0}$.

In contrast to direct semiconductors, in indirect semiconductors, such as silicon or germanium, the conduction band minimum lies far away from the valence band

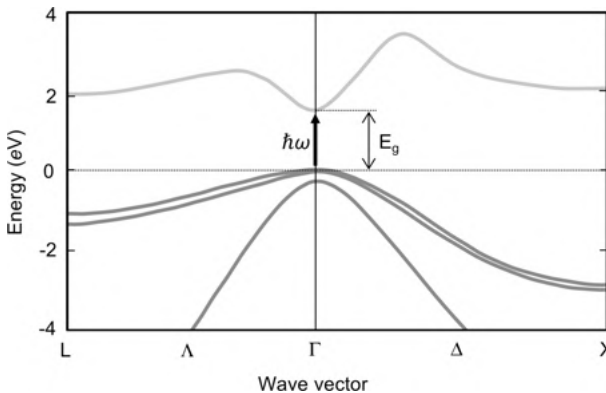


Fig. 2.9: Direct interband absorption at the Γ point of the Brillouin zone in gallium arsenide. For the absorption process to take place at this point, the photon energy $\hbar\omega$ of the excitation light must match the band gap E_g . After Fig. 3.4 of [15], there after [16].

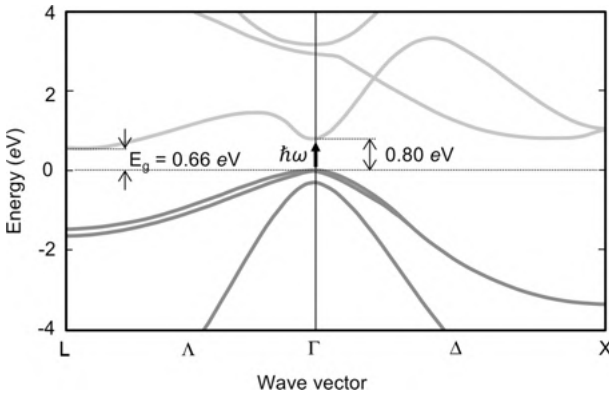


Fig. 2.10: Direct interband absorption at the Γ point of the Brillouin zone in germanium. For the absorption process to take place at this point, the photon energy $\hbar\omega$ of the excitation light must be equal to 0.80 eV, larger than the band gap $E_g = 0.66$ eV. After Fig. 3.10 of [15], there after [17].

maximum, which is at $\vec{k} = \vec{0}$. For instance, the conduction band minimum in germanium lies near the L-point of the Brillouin zone, far away from the Γ -point (Fig. 2.10).

Therefore, in indirect semiconductors, a direct transition is not possible at a photon energy of E_g . In contrast, indirect interband transitions are possible, in which a phonon is additionally absorbed or emitted during the transition. As depicted in Fig. 2.11, the phonon provides the necessary difference in $\vec{k}$ between the initial and the final state of the transition. In an extension of eq. (2.92), the law of momentum conservation for such an indirect interband absorption reads:

$$\hbar\vec{k}_f = \hbar\vec{k}_i \pm \hbar\vec{q}, \quad (2.94)$$

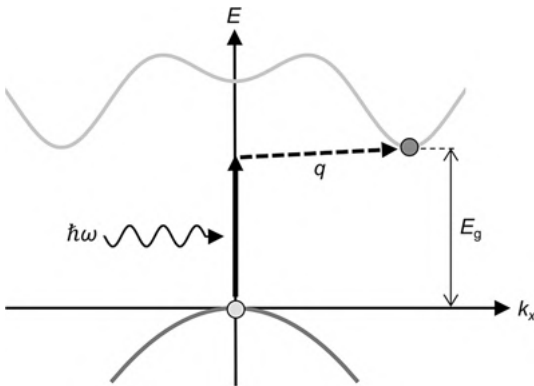


Fig. 2.11: An indirect interband absorption process in an $E(\vec{k})$ diagram. An electron is optically excited from an initial state in an occupied lower band into a final state in an unoccupied upper band. The transition involves a phonon (dashed arrow), which provides the needed momentum $\hbar q$. The energy of the phonon is much smaller than the photon energy.

where $\vec{q}$ is the wave vector of the phonon. As in eq. (2.92), we have neglected the momentum of the photon. The law of energy conservation for the transition reads:

$$E_f = E_i + \hbar\omega \pm \hbar\omega_{\text{ph}} , \quad (2.95)$$

where the + sign stands for the absorption and the – sign for the emission of a phonon. Typical phonon energies $\hbar\omega_{\text{ph}}$ in solids are a few meV to a few tens of meV and, therefore, small in comparison to typical photon energies for interband transitions (several hundred meV to a few eV).

When an electron is excited into a higher band, it leaves behind a positively charged hole in the initial state in the original band. Conceptually, we can think of these two charge carriers as an electron-hole pair. Such an electron-hole pair is electrically neutral. The creation of an electron-hole pair due to the absorption of a photon is called excitation of the electron-hole pair. The total energy acquired by the electron and hole during the excitation is said to be the energy of the electron-hole pair. After the excitation, the electron-hole pair can lose this energy, partially or entirely, due to energy relaxation of the electron and the hole, e.g., by phonon emission. Electron-hole pairs will be discussed in more detail in Subsection 2.3.3.

2.3.1.2 Absorption coefficient for direct interband transitions

According to Subsection 2.2.2.2, the frequency dependence of optical absorption in a medium is controlled by its electronic structure. A special property of crystals is that the electronic energy levels are arranged in continuous bands with forbidden energy gaps between them. The effects of this energetic arrangement on the frequency dependence of the absorption coefficient can be understood from Fig. 2.7. For a transition from the lower band, there exist no suitable final states at photon energies $\hbar\omega$ below E_g . Hence, no absorption takes place, and α is zero at these photon energies. In contrast, transitions between states in the lower and upper band are possible for $\hbar\omega \geq E_g$. Thus, interband absorption sets in at $\hbar\omega = E_g$. The absorption due to interband transitions between states that lie close to the band gap is called band edge absorption. For very large photon energies, no initial and final states lying far enough apart in energy can be found in the two bands, and the absorption vanishes again. Between these limits, the absorption coefficient has a continuous frequency dependence, because the energies of the initial and final states in the bands are continuously distributed.

The goal of this and the following subsections is to derive expressions for the absorption coefficient associated with direct interband transitions. According to eq. (2.16), the absorption coefficient is defined by

$$\alpha = \frac{dI_p}{dz} \left[-\frac{1}{I_p(z)} \right] . \quad (2.96)$$

A light beam can be imagined as a flow of photons that move in the beam direction at the velocity of light. From eqs. (2.6) and (2.79), we obtain the relation between the

intensity of this light and the number N_γ of photons per time t and cross-sectional area A of the beam:

$$I_p = \frac{N_\gamma E_\gamma}{At} . \quad (2.97)$$

The term α in eq. (2.96) thus describes the relative reduction of the photon flow due to absorption in a layer of thickness dz . This relative reduction is proportional to the transition rate $W_{i \rightarrow f}$ of electrons from the electronic ground state to the excited state in the medium:

$$\alpha \propto W_{i \rightarrow f} . \quad (2.98)$$

For a direct transition, the transition rate can be calculated quantum physically using Fermi's Golden Rule:

$$W_{i \rightarrow f} = \frac{2\pi}{\hbar} |\widetilde{M}|^2 g(\hbar\omega) . \quad (2.99)$$

Here $\widetilde{M}$ is the so-called optical matrix element. It describes the effect of the perturbation exerted by the alternating electric field of the light wave on the electrons in the initial states. Moreover, $g(\hbar\omega)$ is the joint density of states, which tells us how many initial and final states that are $\hbar\omega$ apart exist per volume. For the sake of simplicity, we assume that all pairs of states contributing to $g(\hbar\omega)$ have the same matrix elements $\widetilde{M}$, which is reasonable for transitions involving only two bands. In what follows, we will derive expressions for this optical matrix element and the joint density of states. We will insert these expressions into eq. (2.99) in order to calculate the dependence of the absorption coefficient on $\hbar\omega$ using eq. (2.98). The optical matrix element has the form:

$$\begin{aligned} \widetilde{M} &= \langle f | \widehat{H}' | i \rangle = \int \widetilde{\psi}_f^*(\vec{r}, t) \widehat{H}'(\vec{r}, t) \widetilde{\psi}_i(\vec{r}, t) d^3\vec{r} \\ &= e^{\frac{i(E_f - E_i)t}{\hbar}} \int \psi_f^*(\vec{r}) \widehat{H}'(\vec{r}, t) \psi_i(\vec{r}) d^3\vec{r} , \end{aligned} \quad (2.100)$$

where the operator $\widehat{H}'$ describes the perturbation by the alternating electric field of the light wave (eqs. (2.65) and (2.66)), $\widetilde{\psi}_{i,f}(\vec{r}, t)$ are the time-dependent wave functions of the initial and final states, respectively, and $\psi_{i,f}(\vec{r})$ are their time-independent counterparts. The integration is over the entire crystal volume. For the electric field of the light wave, we use the complex notation from eq. (2.9). Inserting this ansatz into eq. (2.100) gives

$$\widetilde{M} = \int \psi_f^*(\vec{r}) e^{\vec{k}_f \cdot \vec{r}} \cdot \vec{E}_0 e^{i\vec{k}\vec{r}} \psi_i(\vec{r}) d^3\vec{r} , \quad (2.101)$$

where we restrict ourselves to the case of a resonant excitation,

$$\omega = \frac{E_f - E_i}{\hbar} . \quad (2.102)$$

The initial and final state of the optical transition are described by Bloch functions (eq. (2.90)). Inserting them into eq. (2.101) and reordering gives:

$$\begin{aligned} \widetilde{M} &= \frac{e}{V} \int u_f^*(\vec{r}) e^{-\vec{k}_f \cdot \vec{r}} \vec{r} \cdot \vec{E}_0 e^{i\vec{k}\vec{r}} u_i(\vec{r}) e^{i\vec{k}_i \cdot \vec{r}} d^3\vec{r} \\ &= \frac{e}{V} \int u_f^*(\vec{r}) \vec{r} \cdot \vec{E}_0 u_i(\vec{r}) \left(e^{-\vec{k}_f \cdot \vec{r}} e^{i(\vec{k}_i + \vec{k}) \cdot \vec{r}} \right) d^3\vec{r} . \end{aligned} \quad (2.103)$$

Taking into account eq. (2.91) and assuming, without loss of generality, that the light is polarized along the x -direction, we obtain

$$\widetilde{M} = \frac{e|\vec{E}_0|}{V} \int u_f^*(\vec{r}) x u_i(\vec{r}) d^3\vec{r} = \frac{N_{uc}|\vec{E}_0|}{V} \int_{\text{unit cell}} u_f^*(\vec{r}) e x u_i(\vec{r}) d^3\vec{r}. \quad (2.104)$$

Here, N_{uc} is the number of unit cells in the crystal and the integration on the right part is over the volume of a single unit cell. The functions $u_i(\vec{r})$ and $u_f(\vec{r})$ are the lattice periodic parts of the Bloch wave functions, which are closely related to the electron orbitals of the individual atoms and are equal in all unit cells. The integral needs to be calculated from the material-specific functions $u_i(\vec{r})$ and $u_f(\vec{r})$. Typically, these functions do not change too strongly within a band. A single matrix element $\widetilde{M}$ is thus a good approximation for all transitions between two bands, which, a posteriori, justifies eq. (2.99). In order to evaluate eq. (2.99), we still need an expression for the joint density of states $g(\hbar\omega)$. Because the initial and final states lie in bands, the joint density of states is a continuous function of $\hbar\omega$. The precise $\hbar\omega$ dependence of this quantity describes how many initial and final states are $\hbar\omega$ apart and additionally obey momentum conservation. It thus depends on the dispersions of the two bands involved in the optical transitions. Here, we will derive it for direct transitions between two isotropic parabolic bands (sketched in Fig. 2.8 and realized, e.g., in the vicinity of the band gap of direct semiconductors). We obtain from eqs. (2.93) and (2.92)

$$\hbar\omega = E_g + \frac{\hbar^2 \vec{k}^2}{2m_e^*} + \frac{\hbar^2 \vec{k}^2}{2m_h^*}, \quad (2.105)$$

where m_e^* and m_h^* are the effective masses of the conduction and valence bands of the direct semiconductor, respectively. Using the reduced mass $\tilde{\mu}$

$$\frac{1}{\tilde{\mu}} = \frac{1}{m_e^*} + \frac{1}{m_h^*}, \quad (2.106)$$

this relation reads

$$\hbar\omega - E_g = \frac{\hbar^2 k^2}{2\tilde{\mu}}. \quad (2.107)$$

This equation tells us that the excess energy $\hbar\omega - E_g$ that the electron-hole pair possesses with respect to the band gap has a quadratic $\vec{k}$ -dependence similar to the dispersion $E(\vec{k})$ of a single parabolic isotropic band. On the basis of this similarity, one can derive an expression for the joint density of states by performing the same steps as in the derivation of the electronic density of states of a parabolic isotropic band (see eq. (1.13) and footnote 16, p. 10). We obtain:

$$g(\hbar\omega) = \frac{1}{2\pi^2} \left(\frac{2\tilde{\mu}}{\hbar^2} \right)^{\frac{3}{2}} (\hbar\omega - E_g)^{\frac{1}{2}} \quad \text{for } \hbar\omega \geq E_g. \quad (2.108)$$

Moreover, it is clear that

$$g(\hbar\omega) = 0 \quad \text{for } \hbar\omega < E_g. \quad (2.109)$$

Inserting these results into eq. (2.99) and making use of eq. (2.98), we obtain the frequency dependence of the absorption coefficient:

$$\alpha(\hbar\omega) \propto (\hbar\omega - E_g)^{\frac{1}{2}} \quad \text{for } \hbar\omega \geq E_g \quad (2.110)$$

$$\alpha(\hbar\omega) = 0 \quad \text{for } \hbar\omega < E_g. \quad (2.111)$$

For a direct semiconductor with isotropic parabolic bands, our treatment thus gives a vanishing absorption coefficient within the band gap (as argued at the start of this subsection) and a square-root dependence on photon energy above the band gap. This prediction is confirmed by experimental measurements. Figure 2.12 shows the absorption coefficient of the direct semiconductor indium arsenide (InAs) at room temperature. Its bands have an isotropic parabolic dispersion in the vicinity of the band gap. The photon energy dependence of α^2 follows a straight line, in agreement with eq. (2.110). The photon energy at which α goes to zero (0.35 eV) represents the band gap energy E_g at room temperature.

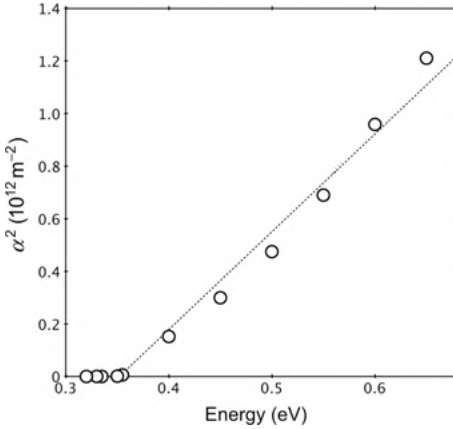


Fig. 2.12: Square of the measured absorption coefficient α versus photon energy $\hbar\omega$ for the direct semiconductor indium arsenide at room temperature. The dashed line is a guide to the eye. After Fig. 3.6 of [15], there after [18].

In many direct semiconductors, the photon energy dependence of the absorption coefficient is found to significantly deviate from eq. (2.110). For instance, at low temperatures and for a sufficiently large band gap, excitonic effects appear, which stem from the Coulomb interaction between the excited electron and hole. The Coulomb interaction enhances the absorption in the spectral vicinity of the band gap. We will consider excitonic effects in Section 2.3.3.1. However, these effects are negligible for the case of room-temperature InAs considered in Fig. 2.12. Moreover, deviations from the parabolic band dispersion that may appear away from the Brillouin zone center may also lead to deviations from the square-root dependence of the absorption coefficient (eq. (2.110)). In such cases, the joint density of states cannot be calculated using an effective mass model, but must be determined numerically on the basis of the real bandstructure.

2.3.1.3 Absorption coefficient for indirect interband transitions

In the previous subsection, we have derived the absorption coefficient for the band-edge absorption in a direct semiconductor. A similar treatment for an indirect semiconductor would be too lengthy to be presented here. It yields the following photon-energy dependence for a valence band maximum at the Γ -point and a conduction band minimum at the L-point with parabolic band dispersions at both band extrema:

$$\alpha(\hbar\omega) \propto (\hbar\omega - E_g \mp \hbar\omega_{\text{ph}})^2, \quad (2.112)$$

where, as in eq. (2.95), $\hbar\omega_{\text{ph}}$ is the phonon energy.

It is clear from eq. (2.112) that the indirect interband absorption must also show a threshold behavior in the spectral vicinity of E_g , albeit shifted by one phonon energy. The sign of the shift depends on whether the phonon involved in the transition is absorbed (–) or emitted (+). Such a shift is indeed observed in Fig. 2.13, which shows data from absorption measurements on the indirect semiconductor germanium. The solid circles represent the square root of the absorption coefficient versus the photon energy in the spectral vicinity of the band-gap energy at 0.66 eV. Above 0.65 eV, the points follow a straight line, as expected from eq. (2.112). This line intersects the photon energy axis at 0.65 eV. Therefore, the absorption sets in already 0.01 eV below the band-gap energy of 0.66 eV. This is attributed to a phonon of energy 0.01 eV, which is absorbed together with the photon. The energy agrees with the energy of the so-called transversal acoustic phonons in germanium. Moreover, the solid circles show an absorption tail below 0.65 eV. This tail can be explained by the absorption of higher-frequency phonons and multiple phonons.

Besides the shift of the absorption threshold, the difference between indirect and direct interband absorption is that the exponent in eq. (2.112) is larger than in

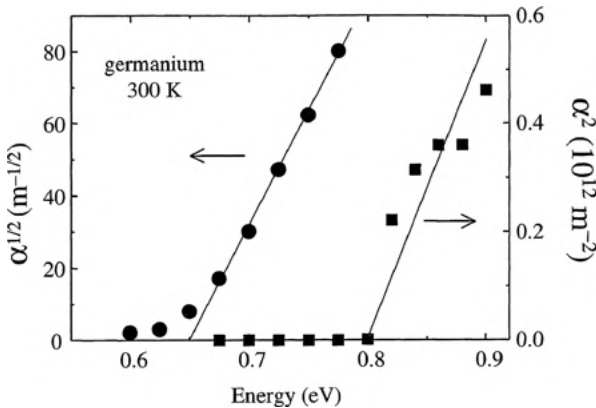


Fig. 2.13: Measured data for the absorption coefficient α of germanium, which is once plotted as $\alpha^{1/2}$ according to eq. (2.112) for indirect transitions and once plotted as α^2 according to eq. (2.110) for direct transitions. Fig. 3.10 of [19], there after [18].

eq. (2.110). The larger exponent is a consequence of the fact that the momentum transfer provided by the phonon enables other combinations of initial and final states than those possible for direct transitions. This difference in absorption can be used to determine whether the band gap of the given semiconductor is of a direct or indirect nature. For instance, a pronounced additional increase of the absorption coefficient is observed for germanium above 0.80 eV. This is shown by the black squares in Fig. 2.13, which exhibit the square of the absorption coefficient versus the photon energy. The increase starting at 0.80 eV follows approximately a straight line and thus obeys the power law of eq. (2.110), which demonstrates that the additional increase is caused by direct absorption. This absorption corresponds to a direct transition at the Γ -point with an energy of 0.80 eV (Fig. 2.10).

Notice that the absolute value of the absorption coefficient above 0.80 eV is orders of magnitude larger than immediately above 0.65 eV. The reason is that an indirect absorption needs a phonon to be absorbed (or emitted) at the same time as the photon is absorbed. The probability for such a transition depends on the probability that these processes occur simultaneously. The process is, hence, a second order process, which has a much lower probability than the first order process of a direct transition, which requires only the absorption of a photon. This difference can also be seen in Fig. 2.14, which shows the measured absorption coefficients of the semiconductors silicon (Si) and gallium arsenide (GaAs). Si possesses an indirect band gap at an energy of 1.12 eV, and GaAs a direct band gap at 1.42 eV. Fig. 2.14 shows that the absorption coefficient of the indirect semiconductor Si at photon energies slightly above 1.1 eV is orders of magnitude smaller than that of the direct semiconductor GaAs at 1.42 eV.

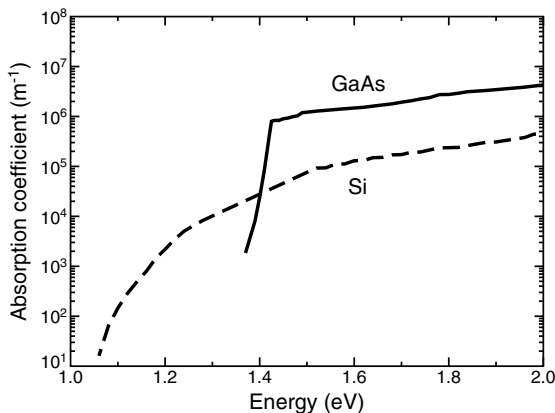


Fig. 2.14: Comparison between the absorption coefficients of the direct semiconductor gallium arsenide and the indirect semiconductor silicon at room temperature. After Fig. 3.9 of [15], there after [18].

2.3.1.4 Absorption spectroscopy

A plot of the absorption coefficient α versus the photon energy $\hbar\omega$ or wavelength λ of the light is called an absorption spectrum. Measuring such spectra and the methods applied in the measurement are called absorption spectroscopy. As we have seen in the context of Fig. 2.13, absorption spectra may reveal valuable information about the optical and electronic properties of a solid. Absorption spectroscopy is therefore a widespread method for the characterization of solids. Experimentally, absorption spectra are taken by measuring, in a spectrally resolved way, the intensity of light transmitted through a thin layer of the investigated material. A typical experimental set-up is shown in Fig. 2.15.

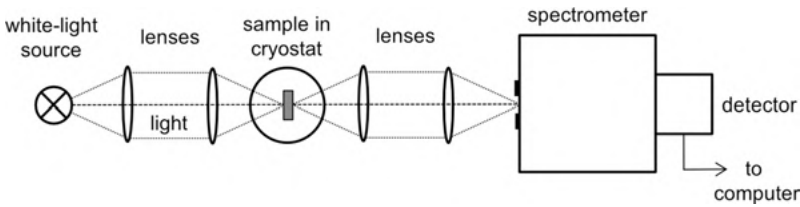


Fig. 2.15: Experimental set up for the measurement of absorption spectra. Light from a white-light source is transmitted through a sample, which is mounted in an optical cryostat for temperature control. The light is spectrally dispersed in a spectrometer and measured with a detector, e.g., a silicon photodiode array detector. The spectrum is recorded with a computer.

Light from a white-light source is transmitted through a sample of known thickness L and spectrally dispersed in a grating spectrometer. The wavelength-dependent light power is measured with a detector, e.g., a silicon photodiode array detector, and recorded with a computer. The ratio of the transmitted light power with and without the sample in the beam path is calculated at each wavelength λ to determine the spectrally resolved transmissivity $T_y(\lambda)$ according to eq. (2.54). The absorption coefficient α is calculated from this data using

$$T_y = (1 - R_{y,1}) e^{-\alpha L} (1 - R_{y,2}) . \quad (2.113)$$

Here, $R_{y,1}$ and $R_{y,2}$ are the reflectivities of the front and rear surface of the sample, respectively, which can be determined in a reflection experiment (Section 2.4.3). The sample can be mounted in an optical cryostat in order to control its temperature. The silicon photodiode array detector is suited to measurements in the sensitivity range of silicon, i.e., roughly $\lambda = 200\text{--}1000$ nm. In the wavelength range from 1000 nm to 1600 nm, it needs to be replaced with an InGaAs array detector. At still greater wavelengths in the infrared, scanning monochromators or Fourier transform spectrometers with suitable infrared detectors can be employed.

2.3.1.5 Semiconductor photodiodes

One of the most important technical applications of interband absorption is the photodiode. This is a semiconductor diode in which photo-generated carriers generate a current that flows in an electric circuit connected to the diode. Photodiodes can be employed as photodetectors and as photovoltaic cells. Figure 2.16 shows the principle of a photodiode that is used as a photodetector, i.e., for the detection of light. It consists of a p-i-n semiconductor structure with an intrinsic, i.e., undoped, layer *i* between a p-doped and an n-doped zone. This results in a band profile where the built-in voltage difference between the p-doped region and the n-doped region drops across the intrinsic zone.⁴ The photodiode is operated using an applied reverse bias voltage V_0 that, hence, adds to the built-in voltage. Consequently, there is a strong electric field in the intrinsic zone, but, in the absence of light, only a very small electric current (dark current). When light with a photon energy above the band gap E_g is incident on the diode, the interband absorption of photons generates free electrons and holes in the *i*-zone. These carriers are pulled in opposite directions by the strong electric field, i.e., electrons are pulled towards the n-zone and holes towards the p-zone. Carriers that reach the contacts are measurable as a photocurrent I_{pc} in the external circuit. This photocurrent can be used to detect light.

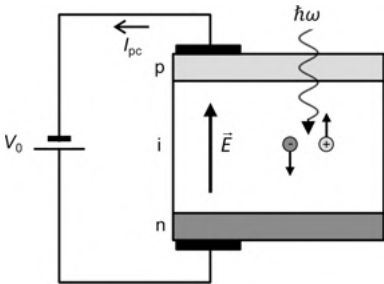


Fig. 2.16: A p-i-n photodiode in an external electric circuit. A reverse bias voltage V_0 is applied to the diode, resulting in an electric field $\vec{E}$ in the intrinsic (*i*) zone. The interband absorption of photons of energy $\hbar\omega$ in the intrinsic zone generates free electrons and holes, which are pulled into opposite directions by the field. Carriers that reach the contact regions *p* and *n* are measurable as a photocurrent I_{pc} in the external circuit.

In what follows, we will estimate the photocurrent. We consider a photodiode whose intrinsic layer has a length L . Light with a photon energy $\hbar\omega > E_g$ and power P is incident on the diode, corresponding to a photon current of $\frac{P}{\hbar\omega}$ photons per second. We assume that the surface of the photodiode is oriented towards the light source and is anti-reflection coated to prevent a reflection of the incident light. Moreover, we assume that the absorption in layers above the intrinsic layer is negligible. According to Beer's law (eq. (2.17)), the portion of the light power absorbed along the distance L is equal to $(1 - e^{-\alpha L})$. Here α is again the absorption coefficient at the photon energy $\hbar\omega$. As each absorbed photon generates one electron-hole pair, of which only one charge carrier contributes to the current in the external circuit, the photocurrent is then given

⁴ Corresponding band profiles can be found, e.g., in [2, Chapter 12.6] or in [15, appendix E].

by

$$I_{pc} = e\eta_q \frac{P}{\hbar\omega} (1 - e^{-\alpha L}) . \quad (2.114)$$

Here, the quantum efficiency η_q denotes the fraction of photogenerated electrons and holes that are extracted into the external circuit. A quantity derived from eq. (2.114) that is important for the suitability of the diode as a photodetector is its responsivity R_{ph} . It is defined as the ratio

$$R_{ph} = \frac{I_{pc}}{P} \quad (2.115)$$

and tells us how efficient the photodiode transforms an incident optical power P into a photocurrent I_{pc} . According to eq. (2.114), the responsivity is

$$R_{ph} = \frac{e\eta_q}{\hbar\omega} (1 - e^{-\alpha L}) . \quad (2.116)$$

For a use of the photodiode as a photodetector, essentially three properties are desirable:

- high responsivity
- low noise
- fast response

According to eq. (2.116), a high absorption coefficient α and a high quantum efficiency η_q are required to achieve a high responsivity of the photodiode. An upper limit to R_{ph} will be reached if both $(1 - e^{-\alpha L})$ and η_q are equal to unity. In this case, the responsivity will be $\frac{e}{\hbar\omega}$. To achieve a high absorption, it is, of course, necessary that the semiconductor material has a band gap $E_g < \hbar\omega$ in the entire spectral region of interest. This criterion makes the choice of the semiconductor dependent on the spectral range to be covered. Because of their higher absorption, direct semiconductors such as gallium arsenide are better suited than indirect ones such as silicon. For many applications, the disadvantage of indirect semiconductors can be compensated by a larger layer thickness, as, e.g., in the case of silicon. For a high responsivity, it is also useful to make the contact that faces the light source as transparent as possible. Therefore, one chooses a small thickness and a suitable material such as, e.g., indium tin oxide (ITO). The second desirable property, i.e., low noise, is achieved by making the band gap E_g as large as possible. This reduces the noisy current contribution that results from carriers thermally excited across the band gap (dark current). For the third property, i.e., a fast response, small layer thicknesses and large carrier mobilities μ of the semiconductor material are advantageous, as they allow a fast extraction of the photogenerated carriers from the photodiode.

Figure 2.17 shows the principle of a photodiode that is used as a photovoltaic cell, i.e., for the transformation of light energy into electric energy. Here, the voltage source from Fig. 2.16 has been replaced by an electric load with resistance R . No photocurrent flows in the dark, meaning that the voltage drop across the load, and thus also the voltage at the contacts of the diode, is zero. When the diode is illuminated, the light

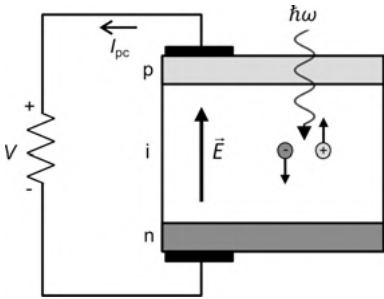


Fig. 2.17: p-i-n photodiode operated as a photovoltaic cell. The photocurrent I_{pc} generated by the illumination results in a power $I_{pc}^2 R$ consumed in the load resistance R marked by its voltage drop V . The voltage $V = I_{pc} R$ dropped across R due to the current flow opposes the built-in field across the intrinsic zone of the diode.

absorption in the intrinsic zone generates electron-hole pairs. The electrons and holes are separated by the built-in field of the diode and swept to the n- and p-zones, respectively. The photocurrent I_{pc} resulting from this motion flows through the load, where it can do work or generate heat. Hence, the photovoltaic cell transforms the absorbed light power into an electrical power RI_{pc}^2 . The voltage RI_{pc} dropped across the load and thus applied to the diode biases the diode in the forward direction. This voltage opposes the built-in field $\vec{E}$ and thus reduces the field transporting the carriers across the intrinsic zone. Therefore, when the illumination power is increased from zero, the photocurrent saturates as soon as $I_{pc}R$ approaches the value of the built-in voltage V_{bi} across the i-zone of the diode. The photocurrent saturation limits the electrical power that can be generated.

2.3.2 Interband luminescence

In Section 2.3.1, we have dealt with the excitation of electrons in solids due to light absorption. The reverse process is known for atoms in which electrons perform transitions from excited levels into lower ones while emitting light (Section 2.2.3). When such a radiative emission process occurs in a solid, it is called luminescence. One distinguishes between photoluminescence and electroluminescence, according to whether the carriers in the initial levels have been excited optically or electrically. Moreover, it is possible to generate stimulated emission (Section 2.2.3.2). This phenomenon is used in semiconductor lasers. In the present section, we will study the features of different kinds of light emission in solids.

2.3.2.1 Optical emission from electron-hole pair recombination

Generally, in luminescence, electrons jump from excited levels into lower ones by spontaneously emitting a photon. As shown schematically in Fig. 2.18, the two levels involved in visible or near-infrared luminescence in solids lie, in general, in two different energy bands. This process is known as interband luminescence. Such a downward transition shares with the upward transitions involved in absorption (Sec-

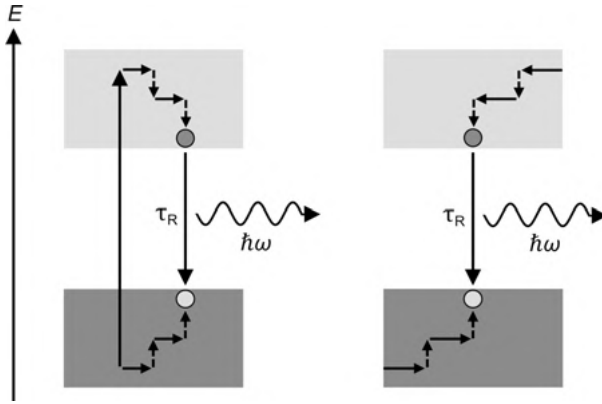


Fig. 2.18: Schematic representations of luminescence processes in a solid in real space. In photoluminescence, carriers are injected into the bands optically (left), and in electroluminescence, electrically (right). In both cases, electrons are injected into the previously empty upper band, while holes are injected into the lower band, which was initially fully occupied with electrons. The carriers rapidly relax into states at or near the band edge via phonon emission (dashed arrows). The electrons subsequently jump into the empty states formed by the holes while emitting photons with energy $\hbar\omega$. τ_R : radiative recombination lifetime.

tion 2.3.1.1) that the initial state must be occupied and the final state empty. The initial occupation is achieved by injecting electrons into states of the previously empty upper band optically, in the case of photoluminescence, or electrically, in the case of electroluminescence. The upper band may be the conduction band of a semiconductor. In general, the injection is into states high up in the band, from where they rapidly relax into states at or near the band edge via phonon emission. In situations with interband luminescence, usually, the lower band is initially fully occupied with electrons, as, e.g., the valence band of semiconductors. Empty states are created in this band by injecting holes simultaneously with the injection of electrons into the upper band. The holes also rapidly relax towards the edge of their band. In the end of the relaxation process, the electron and hole distributions become quasi-stationary. On longer time scales (in general, nanoseconds and longer), the electrons jump into the empty states formed by the holes in the lower band, while emitting photons. This process is called radiative electron-hole recombination.

Let us consider the two levels involved in such a jump. As summarized in Section 2.2.3.2, the spontaneous emission rate in such a two-level system is described by the Einstein A coefficient. If, therefore, the upper level has an occupation N_2 at time t , this occupation decreases at a rate (eq. (2.84))

$$\left(\frac{dN_2}{dt} \right)_R = -A_{21}N_2, \quad (2.117)$$

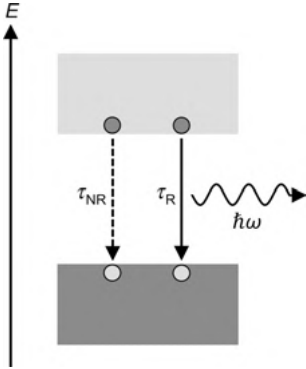


Fig. 2.19: Nonradiative (left) and radiative (right) recombination processes of carriers in band-edge states. τ_R : radiative recombination lifetime, τ_{NR} : nonradiative recombination lifetime.

where „R“ denotes the radiative nature of the occupation change. With the radiative lifetime $\tau_R = A_{21}^{-1}$ (eq. (2.85)), this becomes

$$\left(\frac{dN_2}{dt} \right)_R = -\frac{N_2}{\tau_R}. \quad (2.118)$$

Because the rate at which photons are emitted is $\left| \left(\frac{dN_2}{dt} \right)_R \right|$, eq. (2.118) tells us that the emitted power is proportional to the occupation N_2 of the upper level.

Radiative emission is not the only process by which electrons can make a transition from an excited state into the ground state. The carriers may also lose their excitation energy via emission of phonons, in particular if states within the band gap are available as intermediate states for a sequential phonon emission (Fig. 2.19). Such a nonradiative recombination competes with radiative recombination, i.e., carriers that recombine nonradiatively do not contribute to the emitted light. The total rate of occupation change in the excited level is, hence, obtained by extending eq. (2.118):

$$\left(\frac{dN_2}{dt} \right)_{\text{total}} = -\frac{N_2}{\tau_R} - \frac{N_2}{\tau_{NR}} = -\left(\frac{1}{\tau_R} + \frac{1}{\tau_{NR}} \right) N_2. \quad (2.119)$$

Here, τ_{NR} is called the nonradiative lifetime. The fraction of radiatively recombining electron-hole pairs is called the radiative quantum efficiency η_R :

$$\eta_R = \frac{\left(\frac{dN_2}{dt} \right)_R}{\left(\frac{dN_2}{dt} \right)_{\text{total}}}. \quad (2.120)$$

According to eqs. (2.118) and (2.119), this implies

$$\eta_R = \frac{1}{1 + \frac{\tau_R}{\tau_{NR}}}, \quad (2.121)$$

which is independent of N_2 . If τ_R and τ_{NR} both have identical values for all electron-hole pairs in the vicinity of the band edges, eq. (2.121) holds for all of them and η_R is called the luminescence efficiency or luminescence yield.

If $\tau_{NR} \ll \tau_R$, η_R becomes small and the luminescence is very inefficient. For instance, the luminescence efficiency of the interband luminescence of gold is typically on the order of 10^{-6} , meaning that luminescence is hardly observable. Conversely, an efficient luminescence (i.e., $\eta_R \rightarrow 100\%$) requires that $\tau_{NR} \gg \tau_R$. Direct and indirect transitions behave differently, not only in absorption, but also in luminescence. The law of energy conservation reads for a direct photoluminescent transition, i.e., not involving a phonon:

$$E_f = E_i - \hbar\omega. \quad (2.122)$$

This reflects the fact that the final level is lower than the initial level by the energy of the emitted photon. The corresponding law of momentum conservation reads:

$$\hbar\vec{k}_f = \hbar\vec{k}_i - \hbar\vec{k}_\gamma \approx \hbar\vec{k}_i, \quad (2.123)$$

again exploiting that the momentum of the photon $\hbar\vec{k}_\gamma$ is negligibly small compared to the Brillouin zone (Section 2.3.1.1).

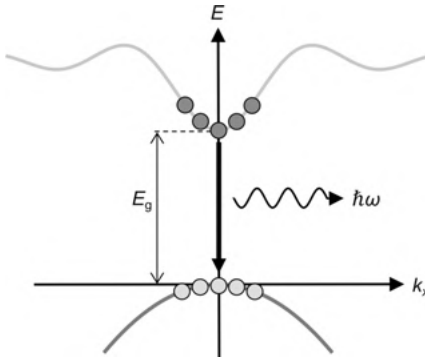


Fig. 2.20: An interband emission process in the $E(\vec{k})$ diagram of a direct semiconductor. An electron jumps from one of the occupied initial states in the conduction band (dark grey circles) across the band gap into one of the empty final states in the valence band (light grey circles) while emitting a photon with energy $\hbar\omega \approx E_g$.

The simultaneous conservation of energy and momentum can again best be illustrated in an $E(\vec{k})$ diagram. For instance, Fig. 2.20 shows a schematic of an interband luminescence in a direct semiconductor such as GaAs. Here, the circles indicate that the lowest part of the conduction band is occupied with electrons, while the topmost part of the valence band is unoccupied (or occupied with holes). These carriers originate from injection into states high up in the respective bands. After their injection, they rapidly relax to the band minima via phonon emission. They form quasistationary distributions, before they recombine radiatively near $\vec{k} = \vec{0}$. Each recombining electron-hole pair hence emits a photon with energy $\hbar\omega \approx E_g$.

This means that independently of how we originally injected the electrons and holes, we obtain luminescence at energies near the band gap. An example is shown in Fig. 2.21. It depicts a photoluminescence spectrum and an absorption spectrum of the direct semiconductor gallium nitride (GaN) at $T = 4$ K. At this temperature, the band gap energy is 3.5 eV. The absorption shows a steep increase close to the band

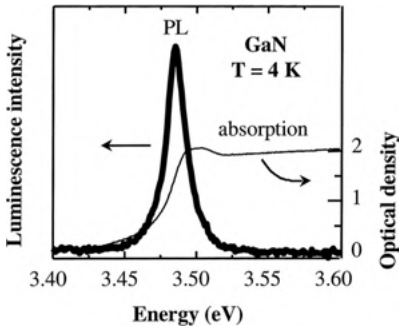


Fig. 2.21: Photoluminescence (PL) spectrum (thick line) and optical density spectrum (thin line) of gallium nitride at $T = 4$ K. The optical density is proportional to the absorption coefficient. The luminescence is excited with light at a photon energy of 4.9 eV.

Fig. 5.3 of [19], from unpublished data from K. S. Kyhm and R. A. Taylor.

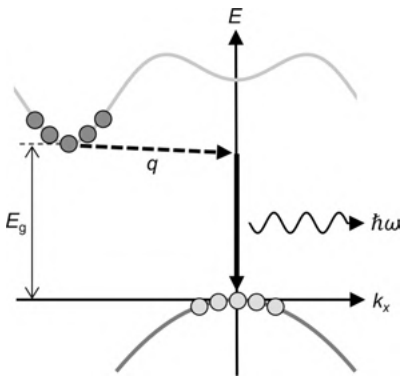


Fig. 2.22: Schematic representation of an interband luminescence process in the $E(\vec{k})$ diagram of an indirect semiconductor. An electron jumps from an occupied initial state in the conduction band across the band gap into an empty final state in the valence band. The transition involves a phonon (dashed arrow), which provides the needed momentum $\hbar q$. The energy of the phonon is much smaller than the photon energy.

gap, which is caused by the so-called Sommerfeld enhancement to be described in Section 2.3.3. Above, the absorption is nearly constant. In contrast, the luminescence consists of a narrow emission line near the band gap energy only.

Figure 2.22 shows a sketch of an interband luminescence in an indirect semiconductor such as silicon. After their injection, the electrons relax into the conduction band minimum away from $\vec{k} = \vec{0}$, and the holes into the valence band maximum at $\vec{k} = \vec{0}$. Then, momentum conservation requires that each emission of a photon is accompanied by the simultaneous emission or absorption of a phonon. The probability for a radiative recombination to occur is determined by the probability of the corresponding second order process, and thus is relatively small. Therefore, the radiative lifetime τ_R is much longer than for direct transitions. According to eq. (2.121), this means that the photoluminescence efficiency η_R is small, as a consequence of the competition with nonradiative recombination processes. For this reason, indirect semiconductors such as silicon and germanium are, in general, poor light emitters.

2.3.2.2 Photoluminescence in solids

In this subsection, we will deal specifically with photoluminescence, which appears after the optical excitation of a solid. Fig. 2.23 shows a schematic drawing of a photoluminescence process in a direct semiconductor such as gallium arsenide. In addition

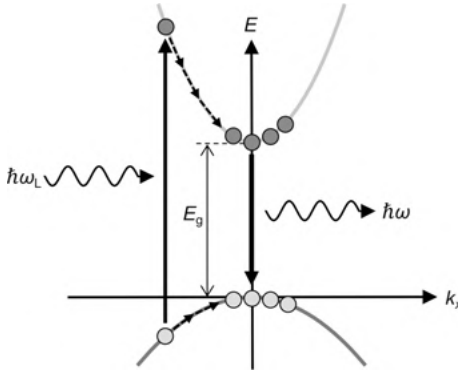


Fig. 2.23: (a) Sketch of photoluminescence process in a direct semiconductor after optical excitation by a laser photon of energy $\hbar\omega_L$. The excitation occurs into high band states, followed by rapid relaxation via sequential inelastic scattering processes (dashed arrows). The carriers accumulate in the vicinity of the band edges and form quasi-stationary thermal distributions, until they recombine radiatively by emission of photons with energy $\hbar\omega$ or non radiatively (not shown).

to the emission process depicted in Fig. 2.20, Fig. 2.23 includes the preceding optical excitation of the electrons and the relaxation of the carriers within both bands. Here, the excitation occurs between states far away from the band extrema, followed by a relaxation via sequential inelastic scattering processes (dashed arrows) by phonons towards the conduction band minimum and the valence band maximum. Each of these scattering events occurs on a timescale on the order of 10^{-13} s. The relaxed carriers accumulate in the vicinity of the band edges and reach thermal equilibrium with the other carriers in the same band via inelastic electron-electron or hole-hole scattering. As a consequence, electrons and holes each form a quasi-stationary thermal distribution, before they recombine radiatively or nonradiatively.

These quasi-stationary thermal distributions can be calculated as follows. The number density of electrons in the conduction band must satisfy the following equation:

$$n_e = \int_0^{\infty} \mathcal{D}_c(E_e) f_e(E_e, T) dE_e, \quad (2.124)$$

where $\mathcal{D}_c(E_e)$ is the density of states and $f_e(E_e, T)$ the Fermi–Dirac distribution function of the excited, quasi-stationary electrons in the conduction band with temperature T . This quasi-stationary distribution is indicated by the subscript e. The electron energy E_e is given with respect to the conduction band minimum. For a parabolic conduction band, $\mathcal{D}_c(E_e)$ reads (eq. (1.13)):

$$\mathcal{D}_c(E_e) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{\frac{3}{2}} E_e^{\frac{1}{2}}. \quad (2.125)$$

The Fermi distribution function is given by

$$f_e(E_e, T) = \left[\exp \left(\frac{E_e - E_{F,e}}{k_B T} \right) + 1 \right]^{-1}. \quad (2.126)$$

Here, $E_{F,e}$ denotes the Fermi energy of the thermal distribution of the excited electrons in the conduction band. This energy is different from the Fermi level of all carriers in

equilibrium. The distinction is important, because the electrons are in thermal equilibrium with each other within the conduction band, but not with the holes in the valence band. Hence, $E_{F,e}$ is called quasi Fermi energy of the electrons. Similarly, the holes are in thermal equilibrium with each other, but not with the electrons. Correspondingly, there is a quasi-Fermi energy $E_{F,h}$ of the holes. The absence of a thermal equilibrium of all carriers with each other, which could be characterized by a single Fermi energy E_F , is a consequence of the fact that there are more electrons and holes after an optical excitation than would be present for a purely thermal excitation across the band gap. Summarizing eqs. (2.124), (2.125), and (2.126), we obtain:

$$n_e = \int_0^{\infty} \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{\frac{3}{2}} E_e^{\frac{1}{2}} \left[\exp \left(\frac{E_e - E_{F,e}}{k_B T} \right) + 1 \right]^{-1} dE_e. \quad (2.127)$$

Analogously, we write for the number density of the holes in the valence band:

$$n_h = \int_0^{\infty} \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2} \right)^{\frac{3}{2}} E_h^{\frac{1}{2}} \left[\exp \left(\frac{E_h - E_{F,h}}{k_B T} \right) + 1 \right]^{-1} dE_h. \quad (2.128)$$

In this integral, we have chosen the maximum of the valence band as the zero of energy, and both E_h and $E_{F,h}$ are counted from the valence band maximum downwards. Because the excitation process generates equal numbers of electrons and holes, $n_e = n_h$ must hold. Both number densities are determined by the intensity of the exciting light. The quasi-Fermi energies can then be determined from a given carrier density n_e or n_h using eqs. (2.127) and (2.128), if T is known. Here, we assume that T is the sample temperature for both electrons and holes.

In the limit of low carrier densities, $E_{F,e}$ lies far below the conduction band minimum and we can neglect the $+1$ term in eq. (2.126). The electron distribution can then be approximated by a Boltzmann distribution function:

$$f_e(E_e, T) \propto \exp \left(\frac{-E_e}{k_B T} \right). \quad (2.129)$$

This also holds for the hole distribution:

$$f_h(E_h, T) \propto \exp \left(\frac{-E_h}{k_B T} \right). \quad (2.130)$$

The probability that a given two-level system with the energy $E_e + E_h = \hbar\omega - E_g$ is occupied both with an electron and a hole is, hence, given by

$$f_e(E_e, T) f_h(E_h, T) \propto \exp \left(\frac{-E_e}{k_B T} \right) \exp \left(\frac{-E_h}{k_B T} \right) = \exp \left(-\frac{\hbar\omega - E_g}{k_B T} \right). \quad (2.131)$$

Multiplying with the joint density of states from eq. (2.108) yields the number density of two-level systems which are occupied both with an electron and a hole. The intensity of the photoluminescence emission is proportional to this number density.

Therefore, the frequency dependence of the photoluminescence intensity for a direct semiconductor reads for $\hbar\omega > E_g$

$$I_p(\omega) \propto (\hbar\omega - E_g)^{\frac{1}{2}} \exp\left(-\frac{\hbar\omega - E_g}{k_B T}\right). \quad (2.132)$$

The behavior can be seen in Fig. 2.24. It shows a photoluminescence spectrum of gallium arsenide at 100 K. The spectrum exhibits a steep rise near E_g due to the density of states factor in eq. (2.132) and then drops exponentially due to the Boltzmann factor. The exponential character of the decay is most easily visible in a semilogarithmic plot of the same data (inset of Fig. 2.24). The slope of the photoluminescence line at high photon energies is in reasonable agreement with the temperature of 100 K at which the measurement was performed. The width (full width at half maximum) of the photoluminescence line is close to $k_B T$, as expected from the combination of steep rise and exponential decay.

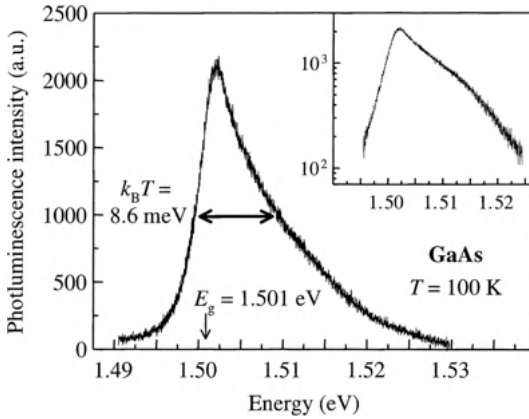


Fig. 2.24: Photoluminescence spectrum of GaAs at a sample temperature of $T = 100$ K. The excitation was performed at a photon energy of 1.96 eV. The inset shows the same spectrum in a semi-logarithmic plot. Fig. 5.6 of [15], there from unpublished data from A. D. Ashmore and M. Hopkinson.

2.3.2.3 Photoluminescence spectroscopy

The measurement of photoluminescence spectra is called photoluminescence spectroscopy. It can give valuable information about the optoelectronic properties of a solid, for instance the energies of states, the existence of defects and the rates of relaxation processes. Photoluminescence spectroscopy is mainly used to characterize semiconductor materials and structures. For instance, it is useful in the development of light-emitting diodes and lasers. Experimentally, the photoluminescence spectra are taken by measuring, in a spectrally resolved way, the intensity of light that has been emitted by a sample after an optical excitation at a high photon energy. A typical

experimental setup is shown in Fig. 2.25. The sample is mounted in a cryostat, which makes it possible to control its temperature. The photoluminescence is excited with laser or lamp light at a photon energy above the band gap. A portion of the photoluminescence light emitted by the sample is collected with a lens and guided to a grating spectrometer, in which it is dispersed spectrally. The wavelength dependent light power is measured with an array detector, e.g., a CCD array. Alternatively, the spectrometer can be replaced with a monochromator and the intensity is measured separately at each wavelength with a more sensitive detector, e.g., a photomultiplier tube. In both cases, the measured data is read out with a computer and saved for evaluation.

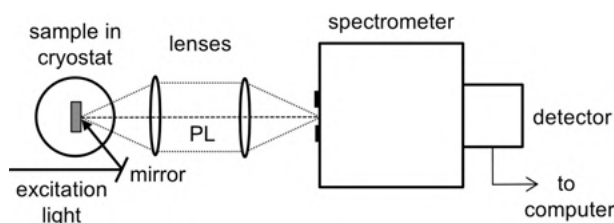


Fig. 2.25: Experimental setup for the measurement of photoluminescence spectra. The sample, which is mounted in a variable temperature cryostat, is excited with light at a photon energy above the band gap of the sample material. The photoluminescence spectrum is obtained by measuring the emission (PL) of the sample as a function of wavelength using a spectrometer and a detector.

In a version of this spectroscopy method, called photoluminescence excitation spectroscopy (PLE), the luminescence intensity is measured at the wavelength of the emission maximum, while the wavelength of the excitation light is varied. The spectra consist of plots of the luminescence intensity versus the photon energy of the exciting light. A feature of this measurement is that the signal is proportional to the density of excited carriers, which, in turn, is proportional to the absorption coefficient at the excitation wavelength. Therefore, the technique can be used to determine the absorption coefficient of samples which are too thick for transmission measurements.

The two varieties of photoluminescence spectroscopy presented here provide a spectral, but not a temporal resolution. The latter can be achieved if the luminescence intensity at the selected wavelength is measured as a function of time after the absorption of a short laser pulse which excites the sample. The measurements are taken using the setup depicted in Fig. 2.25, but with a pulsed laser as the light source and a monochromator as the spectrometer. The time resolution is limited by the pulse duration and the response time of the detection technique. Time resolutions down to 0.1 picoseconds (10^{-13} s) are possible. The time dependence of the measured photoluminescence transients enables the determination of radiative lifetimes and yields information about the dynamics and efficiencies of carrier relaxation and recombination processes.

2.3.2.4 Electroluminescence

Electroluminescence is luminescence generated by electrical injection of electrons and holes into the crystal. Two optoelectronic devices based on electroluminescence, the light-emitting diode (LED) and the laser diode, have found important technical applications. We will study these two devices in this subsection.

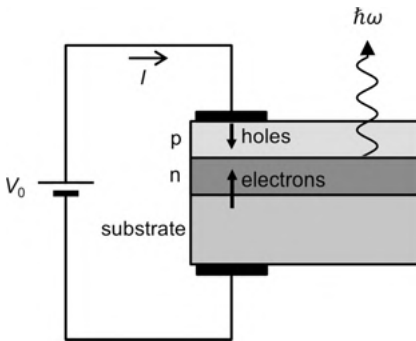


Fig. 2.26: Sketch of a light-emitting diode (LED). The diode is operated in the forward direction, which gives rise to a strong current flow through the layers. The current is composed of carriers that are injected into the transition region between the n- and p-layers. In this zone, the electrons recombine with the holes, emitting photons ($\hbar\omega$) via interband luminescence. The recombining carriers are continuously replaced by the current I that flows through the diode from the external circuit.

Figure 2.26 shows the principle of an LED. It is composed of various thin semiconductor layers, which have been deposited on a thick crystalline semiconductor substrate (Section 1.3.1). The task of the substrate is to support the thin layers above it. These layers form a p-n diode or p-i-n diode. By connecting it to a voltage source, the diode is operated in forward direction, which gives rise to a strong current flow through the layers. The current is composed of carriers that are injected into the transition region between the two layers, the active zone. The electrons are injected through the n-layer, and the holes through the p-layer. In the active zone, the electrons recombine with the holes, emitting photons with energy E_g via interband luminescence. The recombining carriers are continuously replaced by the current that flows through the diode from the external circuit as depicted in Fig. 2.26.

Figure 2.27 shows the electroluminescence spectrum of a p-i-n diode made from gallium arsenide and biased in the forward direction. Light is generated in the thin intrinsic layer between the p- and n-layers. The emission line is at a photon energy of 1.42 eV, which is the band gap energy of GaAs at room temperature. The width of the

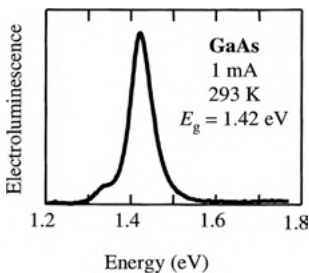


Fig. 2.27: Room-temperature electroluminescence spectrum of a light-emitting diode made from gallium arsenide. Fig. 5.13 of [15], there from unpublished data from A. D. Ashmore.

line (full width at half maximum) is close to 60 meV. This approximately corresponds to twice $k_B T$ and thus is in the order of magnitude expected from the discussion of photoluminescence linewidths in Section 2.3.2.2 (eq. (2.132)). LEDs have found widespread use as light sources for lighting, displays, signals and data communication.

Not only LEDs, but also lasers can be built from semiconductor diodes. They are called laser diodes or semiconductor lasers. For this purpose, the diode structure must be modified in two respects (Section 2.2.3.2). First, a high photon density is required for stimulated emission. Therefore, a laser resonator with appropriate mirrors must be incorporated. This can be achieved by increasing the reflectivities R_y of the semiconductor-air interfaces at the crystal edges (typically $R_y \approx 30\%$) by suitable coatings. Second, an inversion must be generated in the semiconductor medium. This is achieved by injecting strong currents I_{in} , which make the carrier density in the active zone high. The gain coefficient g_ω (eq. (2.86)) then reaches high values. If we assume that g_ω grows linearly with I_{in} , as shown in Fig. 2.28(a), there will be a value of I_{in} at which the gain reaches a large enough value of g_ω , as determined by eq. (2.89), and laser emission sets in. This value is called the threshold current I_{th} .

For all $I_{in} \geq I_{th}$, the value of g_ω is fixed by the equilibrium condition in eq. (2.88), and thus independent of I_{in} (an effect called gain clamping). At currents beyond the threshold current, the additional electrons and holes injected into the active zone cause no additional gain, but give rise to a stronger light field, which is amplified by the same factor per length due to the correspondingly larger density of electron-hole pairs. Above the laser threshold, the output power P_{out} then grows according to $P_{out} \propto (I_{in} - I_{th})$, as depicted in Fig. 2.28(b). If η_L denotes the fraction of injected electrons that emit laser photons, one gets:

$$P_{out} = \eta_L \frac{\hbar\omega}{e} (I_{in} - I_{th}) . \quad (2.133)$$

In an ideal laser diode η_L would be unity, i.e., each electron passing through the device emits a photon. The best laser diodes get very close to this value.

As semiconductor lasers have, due to the built-in resonator, a more complex structure than LEDs, their fabrication is more expensive. However, they have important advantages for applications, such as narrower linewidths, higher output efficiencies, better beam quality and higher response speeds than LEDs. They have found widespread use as light sources for various applications, e.g., fiber optics communication, laser pointers, CD and DVD players, barcode scanners, laser printers and laser surgery.

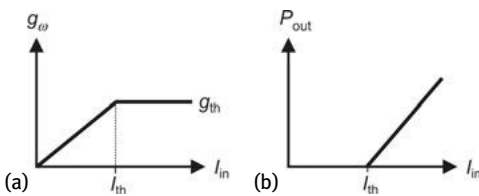


Fig. 2.28: Behavior of a semiconductor laser diode in dependence of injection current I_{in} . Gain g_ω (a) and output power P_{out} (b) versus injection current I_{in} . I_{th} is the threshold current, and g_{th} the threshold gain. After Fig. 5.15 of [15].

Commercial LEDs and laser diodes are fabricated from direct band-gap semiconductors such as GaAs, because the radiative efficiency of materials with an indirect band gap is too low. The choice of direct semiconductors is mainly guided by the following criteria:

- size of the band gap and, hence, emission frequency ω .
- restrictions due to the lattice matching between the semiconductor layers.
- ability to dope the semiconductor material.

The size of the band gap is important as it determines the emission wavelength. The term lattice matching refers to the necessity of matching the lattice constants of subsequent semiconductor layers of the device as well as possible. This is necessary to avoid dislocations due to distorting forces between the lattices (Section 1.3.1). Such dislocations typically reduce the optical quality by enabling non-radiative recombination and decrease the electrical quality through a reduction of the conductivity due to additional scattering. The doping of the semiconductor materials is required to enable the production of the n- and p-layers of the diode.

2.3.3 Excitons

Absorption and photoluminescence spectra of semiconductors at low temperatures show lines in the spectral vicinity of the band gap that are not predicted by the simple picture of interband transitions introduced above. These lines can be understood from the following considerations. When an electron-hole pair is created, e.g., as a consequence of an interband absorption, the electron and hole attract each other via the Coulomb interaction. In certain conditions, this attraction is able to bind the electron and hole to each other. The resulting carrier pair looks electrically neutral to its environment and is called an exciton. Quantum physically, it represents a quasiparticle like a phonon. Its lifetime is finite, because, after its excitation, it can either decay by emitting a photon or it may get destroyed by scattering processes that separate the electron and the hole. Excitons are observed in low temperature absorption and photoluminescence spectra of semiconductors and insulators. In principle, they can also be created in metals; however, the carrier scattering and the screening are too strong to preserve the excitons for a relevant period of time. Essentially, there are two kinds of excitons: Wannier–Mott excitons and Frenkel excitons. In this section, we will study the properties of both kinds in various materials and for different excitation conditions.

2.3.3.1 Wannier–Mott excitons

Wannier–Mott excitons typically appear in direct semiconductors at low temperatures. An important property of these excitons is that the mean distance between electron and hole is much larger than the lattice constant of the crystal, as shown schematically in Fig. 2.29.

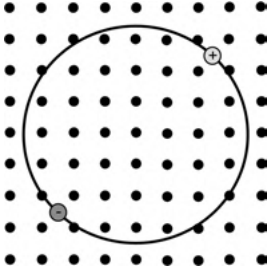


Fig. 2.29: Schematic representation of a Wannier-Mott exciton in a crystal. Black dots represent the atoms. The relative motion of the electron (-) and hole (+) is classically pictured as a common orbit around their shared center of mass. After Fig. 4.1 of [15].

The mean electron-hole distance is thus orders of magnitude larger than the mean distance between the electron and the proton in a hydrogen atom. The difference is a consequence of field screening. The Coulomb attraction force, which binds the negatively charged electron and the positively charged hole, also generates polarization charges in the surrounding medium, which, in turn, partially screen the Coulomb fields of the electron and hole, i.e., partially balance their attraction force. Therefore, the electron and hole are bound less tightly than the electron and proton in the hydrogen atom.

Because the distance between electron and hole is so large, we can average over the atomic structure of the lattice and describe the field screening by the position-independent dielectric constant ϵ (Section 2.2.2.2). In particular, the Coulomb attraction force in the crystal is of a similar $\frac{1}{r^2}$ form as in vacuum. Therefore, the Schrödinger equation for the two-particle wavefunction $\Psi(\vec{r}_e, \vec{r}_h)$ of the electron-hole pair has the same form as for the electron-proton pair of the hydrogen atom – with the exception that the Coulomb interaction term in the Hamilton operator is a factor of $\frac{1}{\epsilon}$ smaller and that we use as particle masses the effective masses m_e^* and m_h^* of electron and hole (Section 1.2.2, eq. (1.6)):

$$\left[\frac{\hat{p}_e^2}{m_e^*} + \frac{\hat{p}_h^2}{m_h^*} - \frac{e^2}{4\pi\epsilon_0\epsilon|\vec{r}_e - \vec{r}_h|} \right] \Psi = E\Psi, \quad (2.134)$$

where $\hat{p}_e$ and $\hat{p}_h$ are the momentum operators for the electron and hole, respectively. In this form of the Schrödinger equation, we assume that the electron-hole pair already exists and only the kinetic energies of both carriers as well as the interaction between them must be included – precisely as in the case of the hydrogen atom, in which the electron and proton already exist before their mutual binding. This means that we will have to consider, a posteriori, the energy invested for the creation of the electron-hole pair. When we use the separation ansatz

$$\Psi(\vec{r}_e, \vec{r}_h) = F(\vec{R})G(\vec{r}), \quad (2.135)$$

rewriting the two-particle wave function in terms of the center-of-mass coordinate

$$\vec{R} = \frac{m_e^* \vec{r}_e + m_h^* \vec{r}_h}{m_e^* + m_h^*} \quad (2.136)$$

and the relative coordinate

$$\vec{r} = \vec{r}_e - \vec{r}_h, \quad (2.137)$$

the Schrödinger equation breaks up – as in the case of the hydrogen atom – into an equation for the center-of-mass motion and an equation for the relative motion, which can be solved separately. As part of the solution for the center-of-mass motion, we obtain the energy eigenvalues

$$E^{\text{center}} = \frac{\hbar^2 \vec{K}^2}{2(m_e^* + m_h^*)}, \quad (2.138)$$

where $\hbar \vec{K}$ is the center-of-mass momentum of the exciton. The quadratic energy dispersion appearing in eq. (2.138) is characteristic of free particles (Section 1.2.2) and shows that the Wannier–Mott exciton can freely move as a unit through the crystal. Therefore, it is also called a free exciton. As part of the solution for the relative motion we obtain energy eigenvalues of an analogous form as in the case of the hydrogen atom:

$$E^{\text{rel}} = -\frac{E_{\text{Ryd},X}}{n^2}, \quad (2.139)$$

where

$$E_{\text{Ryd},X} = \frac{\tilde{\mu}}{m_e \epsilon^2} E_{\text{Ryd}} \quad (2.140)$$

is known as the excitonic Rydberg constant or excitonic binding energy. The principal quantum number is called n , $\tilde{\mu}$ is the reduced mass of the electron-hole pair (eq. (2.106)), m_e the electron mass in vacuum, and $E_{\text{Ryd}} = 13.6 \text{ eV}$ the Rydberg constant of the hydrogen atom. The negative sign in eq. (2.139) indicates that the eigenfunctions of the relative motion are bound states, just like the bound electron states in the hydrogen atom. In a classical picture, the relative motion of an electron and hole can be imagined as an orbit around their shared center of mass, as illustrated in Fig. 2.29. The radius of this orbit is given by

$$r_n = n^2 a_X, \quad (2.141)$$

where

$$a_X = \frac{m_e \epsilon}{\tilde{\mu}} a_B \quad (2.142)$$

is called the exciton Bohr radius and $a_B = 0.5 \text{ \AA}$ is the Bohr radius of the H atom. According to eqs. (2.139) and (2.141), the $n = 1$ state has the largest binding energy and the smallest orbital radius. The higher states with $n > 1$ have smaller binding energies and larger orbital radii. The possible energies that the exciton can have are consequently given as a sum of the above energies:

$$E_{n,\vec{K}} = E_g + \frac{\hbar^2 \vec{K}^2}{2(m_e^* + m_h^*)} - \frac{E_{\text{Ryd},X}}{n^2}. \quad (2.143)$$

Here, the first term takes into account that an excitation across the band gap of the direct semiconductor is required for an electron-hole pair to be created. This energy term was not included in eq. (2.134). The second term represents the energy eigenvalues of the center-of-mass motion, and the third one those of the bound states of the relative motion. The energy-level scheme described by eq. (2.143) is shown in Fig. 2.30. Here, the energies of the bound electron-hole states are depicted as parabolas. The continuously distributed energy levels of the unbound electron-hole states (shaded gray) are denoted as an electron-hole continuum. In a classical picture, these states can be imagined as non-closed trajectories, in which the electron and hole pass each other with so much kinetic energy that they cannot trap each other into a bound state. Similarly to the bound states, these states are solutions to the Schrödinger equation (eq. (2.134)), but have energies exceeding those described by eq. (2.139) due to the kinetic energies of unbound electrons and holes.

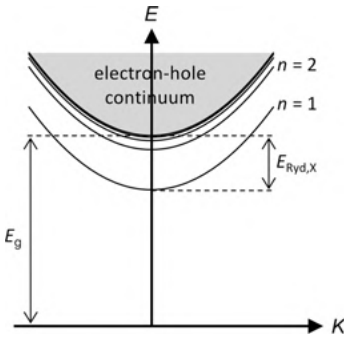


Fig. 2.30: Energy level scheme of Wannier-Mott excitons according to eq. (2.143). After Fig. 4.1 of [15].

The optical excitation of excitons is subject to energy and momentum conservation. This means that an exciton that is in the process of being created takes up the energy and momentum of the absorbed photon. Hence, only those exciton states can be excited which lie at intersections of the level scheme in Fig. 2.30 with the $E(\vec{k})$ dispersion relation of light in the solid (eq. (2.5)):

$$\hbar\omega = \hbar c|\vec{k}|. \quad (2.144)$$

However, as discussed in Section 2.3.1.1, the photon wave number $|\vec{k}|$ in the visible range is much smaller than the size of the Brillouin zone, which implies transitions at $\vec{K} \approx \vec{0}$ in Fig. 2.30. We thus may simplify eq. (2.143) to

$$E_n = E_g - \frac{E_{\text{Ryd},X}}{n^2}. \quad (2.145)$$

We therefore expect absorption lines at the photon energies given by eq. (2.145). An example of such an absorption is given in Fig. 2.31, which shows a measured absorption spectrum of gallium arsenide at low temperature (1.2 K). The three lowest

excitonic lines ($n = 1, n = 2, n = 3$) are visible. As expected from eq. (2.145), the two lowest lines are the furthest apart. The higher ones lie increasingly closer, such that lines with $n > 3$ can not be resolved due to their lifetime broadenings. Above $E_g = 1.5191$ eV, the absorption is caused by the excitation of unbound electron-hole pair states (electron-hole continuum). Immediately above E_g , much higher values of α are observed than those predicted by eq. (2.110). This enhancement is called the Sommerfeld factor. It is a consequence of the enhanced overlap of the electron and hole wave functions. This overlap is caused by the Coulomb attraction force between the two carriers. The overlap enhances the transition probability compared to independent carriers via the matrix element $\widetilde{M}$ (eqs. (2.103)). This enhancement is also visible in Fig. 2.21.

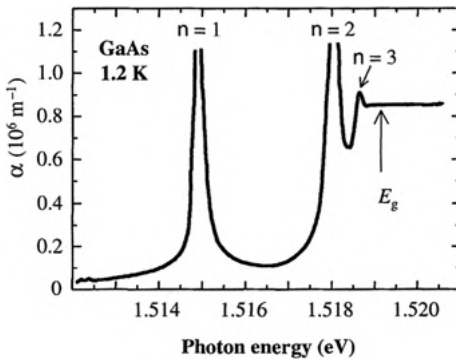


Fig. 2.31: Measured absorption spectrum of GaAs at a temperature of 1.2 K. The labels mark the three lowest excitonic absorption lines ($n = 1, n = 2, n = 3$) and the energy of the band gap E_g . Fig. 4.4 of [19], there after [20].

According to eq. (2.145), the exciton binding energy in gallium arsenide can be determined from Fig. 2.31 as the difference between the energy of the ($n = 1$)-line (1.5149 eV) and the band gap energy (1.5191 eV) revealing $E_{\text{Ryd},X} = 4.2$ meV. Whether any exciton lines can be observed in a given semiconductor at a given temperature depends on this exciton binding energy. Stable excitons only exist if the binding energy is large enough to preserve the bound state from destructive collisions with phonons. If the temperature rises so that the thermal energy $k_B T$ increases beyond $E_{\text{Ryd},X}$, it becomes likely that the electron-hole pair is excited to an unbound state by phonon absorption, and is, hence, dissociated. Consequently, the exciton line in the spectrum disappears as shown in Fig. 2.32. This process is called thermal ionization of the exciton. Because the electron and hole in a Wannier–Mott exciton attract each other relatively weakly due to the high ϵ , Wannier–Mott excitons have small binding energies on the order of 10 meV. As $k_B T$ is 25 meV at room temperature, exciton lines in direct semiconductors are observed only at low temperatures. In contrast to the lines themselves, the Sommerfeld enhancement is more persistent, appearing even at room temperature, as shown in Fig. 2.32.

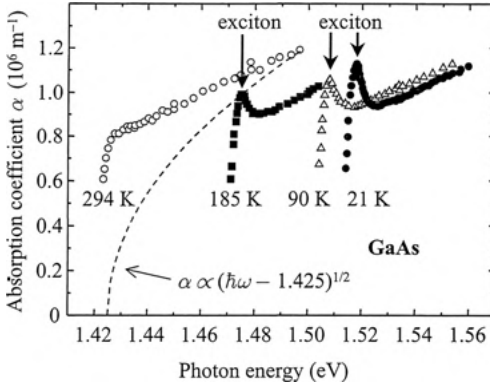


Fig. 2.32: Measured absorption spectra of gallium arsenide at different temperatures. In addition to a band edge shift typical of semiconductors and caused by the thermal expansion of the crystal, the spectra show an increasing thermal ionization of the excitons with increased temperature via the disappearance of the exciton line. The dashed line shows the result of a fit to the measured spectrum at 294 K using eq. (2.110). At this temperature, the band-gap energy is $E_g = 1.425$ eV. Fig. 4.3 of [15], there after [21].

Excitons may also get ionized as a consequence of an external electric field. When a strong, time-independent, electric field acts on an exciton, the electron and hole are pulled apart. It is possible to show that the field strength required for this phenomenon is

$$|\vec{E}_{\text{critical}}| \approx \frac{2E_{\text{Ryd},X}}{ea_X}. \quad (2.146)$$

This exciton dissociation is called field ionization of the exciton. It can be observed in absorption spectra as a lifetime broadening of the exciton ($n = 1$)-line or as a complete disappearance of the line. Thermal and field ionizations both contribute to the fact that excitonic effects only play a minor role in most semiconductor optoelectronic devices. The value of the binding energy $E_{\text{Ryd},X}$ is critical for the stability of the exciton against thermal and field ionizations. A systematic comparison of exciton binding energies for direct semiconductors shows that $E_{\text{Ryd},X}$ generally grows with increased band gap. The reason is that the effective masses of the free carriers increase with increasing E_g and that polarization charges are less easily induced at large E_g . Consequently, the reduced mass $\tilde{\mu}$ of the electron-hole pairs becomes larger and the dielectric constant ϵ smaller. Both trends increase the exciton binding energy (eq. 2.140). Additionally, the exciton Bohr radius a_X decreases with increasing E_g (eq. (2.142)). In insulators with band gaps above 5 eV, a_X is similar to the lattice constant. Consequently, the model of the Wannier–Mott exciton loses its meaning. Conversely, in semiconductors with small band gaps such as indium arsenide ($E_g = 0.35$ eV), it is difficult to observe any effects of Wannier–Mott excitons due to ionization, as shown in Fig. 2.12. Therefore, Wannier–Mott excitons can best be observed in semiconductors with medium-sized band gaps (1 eV to 3 eV).

2.3.3.2 Frenkel excitons

Frenkel excitons appear in insulators with large E_g , in molecular crystals and conjugated polymers. A characteristic property of these excitons is that the mean distance between electron and hole is similar to the lattice constant (Fig. 2.33). As explained

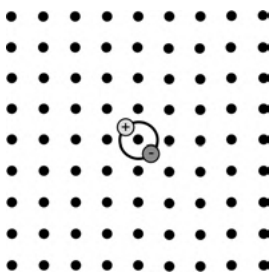


Fig. 2.33: Schematic representation of a Frenkel exciton in a crystal. The relative motion of the electron (e) and hole (h) is pictured as a common orbit of electron and hole around a single atom or molecule of the crystal.

in the previous subsection, a large E_g implies a weak screening of the Coulomb interaction between electron and hole resulting in a strong attraction. Frenkel excitons, hence, have small radii and large binding energies, which typically range from 0.1 eV to several eV. Their binding energies are thus much larger than the thermal energy $k_B T$, and the Frenkel excitons are normally stable at room temperature. Large band gaps are also associated with large effective masses of the carriers. This explains why Frenkel excitons are much less mobile than Wannier–Mott excitons. They are bound to single atoms or molecules of the crystal lattice and can only move by phonon-assisted transport between lattice sites, a process called exciton hopping (see also Section 1.6.3).

For the theoretical treatment of Frenkel excitons, there is no simple model. The calculation usually follows a tight-binding approach, which emphasizes the analogy to atomic and molecular states.

A class of materials in which Frenkel excitons have been demonstrated are the alkali halogenide crystals, e.g., sodium chloride. They have large band gaps, i.e., their excitonic lines lie in the ultraviolet. Their excitonic binding energies are between 0.3 eV for sodium iodide and 0.9 eV for lithium fluoride. These values are far larger

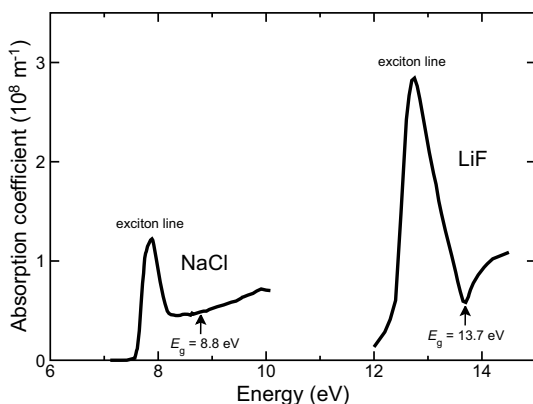


Fig. 2.34: Absorption spectra of sodium chloride and lithium fluoride at room temperature. The exciton lines and the band gap absorption at E_g are marked. After Fig. 4.8 of [15], there after [18].

than $k_B T = 25$ meV at room temperature. Therefore, the exciton lines do not disappear at this temperature, in contrast to those in gallium arsenide. Figure 2.34 shows absorption spectra of sodium chloride and lithium fluoride. From the energy differences between the ($n = 1$) exciton lines and the band gaps E_g , the excitonic binding energies are determined to be $E_{\text{Ryd},X} = 0.8$ eV for sodium chloride and $E_{\text{Ryd},X} = 0.9$ eV for lithium fluoride.

2.4 Free-electron excitations in solids

The electric field of a light wave is able to induce interband transitions as described in the previous sections, but also transitions between states within the same band, so-called intraband transitions. Such intraband transitions are known from electric transport, where they describe the transition of electrons between states of different crystal momentum $\hbar\vec{k}$ within the same band, corresponding to a change of the velocity of the electrons in real space (Section 1.2.2, Fig. 1.3). Without scattering, the instantaneous crystal momentum of the electron at time t is given by Bloch's acceleration theorem

$$\hbar\vec{k}(t) = \hbar\vec{k}(t=0) - e \int_0^t \vec{E}(\tilde{t}) d\tilde{t}. \quad (2.147)$$

The corresponding momentum changes are, however, minute at the high frequencies ω of the light, which imply a fast change of the electric field $\vec{E}(t)$. Nevertheless, they have non-negligible effects if a large number of electrons occupy the same band, e.g., the conduction band of a doped semiconductor or of a metal. The collective motion resulting from the momentum changes may lead to a considerable electrical current. If the frequency of the driving electric field is in the optical range, the rapidly oscillating current will influence the optical properties of the crystal. In the present section, we will study this influence in electrical conductors such as metals.

2.4.1 Optical properties of metals

As an example of a good electrical conductor, we consider a typical non-ferromagnetic metal, e.g., silver. Here, we base most of our treatment on methods from classical physics. The time-dependent electric and magnetic fields at the surface and in the interior of the metal are described by the Maxwell equations (eqs. (2.41)–(2.44)). In contrast to a dielectric, a conductor such as a metal possesses free electrons, which are not bound to individual atoms but are able to move freely through the medium and contribute to electrical currents. Therefore, the electrical current densities $\vec{j}(t)$ induced in the medium by the alternating electric field of the light wave are non-negligible. Unlike in Section 2.2.2.2, we are thus not allowed to set $\vec{j}$ to zero. In analogy to eq. (2.46),

we obtain from the Maxwell equations:

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \varepsilon_d \frac{\partial^2 \vec{E}}{\partial t^2} + \mu_0 \frac{\partial \vec{j}}{\partial t}, \quad (2.148)$$

where ε_d is the dielectric constant that the material would have without electric currents. It describes how far the bound charges in the medium can be displaced by the electric field. Inserting into eq. (2.148) the expression from eq. (1.1)

$$\vec{j} = \sigma \vec{E} \quad (2.149)$$

yields

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \varepsilon_d \frac{\partial^2 \vec{E}}{\partial t^2} + \mu_0 \sigma \frac{\partial \vec{E}}{\partial t}. \quad (2.150)$$

This extends the wave equation, eq. (2.46), by the electrical conductivity σ . Inserting the ansatz of a plane harmonic wave according to eq. (2.26), we obtain

$$\tilde{k}^2 = \varepsilon_d \frac{\omega^2}{c_0^2} + i \mu_0 \sigma \omega. \quad (2.151)$$

A comparison with eq. (2.25) and use of $c_0 = 1/\sqrt{\varepsilon_0 \mu_0}$ yields for the complex refractive index

$$\tilde{n}^2 = \varepsilon_d + i \frac{\sigma}{\varepsilon_0 \omega}, \quad (2.152)$$

which extends eq. (2.48) by a conductivity term. Equation (2.152) reveals that the macroscopic optical properties of the medium described by $\tilde{n}$ (phase velocity and absorption) are determined by two microscopic properties, namely the polarizability of the bound charges (described by ε_d) and the ability to conduct electrical currents (described by σ). In the frequency range of light, σ differs from its value in the direct current case ($\omega = 0$). To emphasize this difference, $\sigma(\omega)$ is often called optical conductivity for frequencies in the visible range. To derive an expression for its frequency dependence, we consider each of the conduction electrons as a classical free point-like particle with mass m_e^* and charge $-e$ that moves under the effect of the periodic force $-e\tilde{E}(\vec{r}, t) = -e\tilde{E}_0 e^{i(\vec{k}\cdot\vec{r} - \omega t)}$ and is subject to a frictional force due to scattering at a rate γ . This approach is inspired by the Drude model, which describes the electric transport of free electrons in metals (Section 1.2.2). The equation of motion for the field-induced electron displacement $\vec{x}$ from its initial position reads

$$m_e^* \ddot{\vec{x}} + m_e^* \gamma \dot{\vec{x}} = -e\tilde{E}(\vec{r}, t). \quad (2.153)$$

This equation differs from the equation of motion for the Lorentz oscillator (eq. (2.11)) in that the restoring force of the oscillation is absent. The reason is that the electrons considered here are conduction electrons of the metal, which are not bound to individual atoms, but are able to move freely through the medium. Equation (2.153) can

also be expressed by the velocity $\vec{v} = \dot{\vec{x}}$ of the electron (see also eq. 1.5):⁵

$$m_e^* \dot{\vec{v}} + m_e^* \gamma \vec{v} = -e \vec{E}(\vec{r}, t) . \quad (2.154)$$

The term $\vec{v}$, in turn, can be expressed by the electric field $\vec{E}$ in the following way. The relation between the electric current density $\vec{j}$ and the number N_e of electrons flowing through the cross-sectional area A of the conductor per time t is (eq. (1.2)):

$$|\vec{j}| = \frac{N_e e}{At} = \frac{N_e}{V} (-e) |\vec{v}| = -n_e e |\vec{v}| , \quad (2.155)$$

where $V = A|\vec{v}|t$ is the volume filled by the N_e electrons contributing to the current during time t and, hence, n_e is the electron density. Combining eq. (2.155) with eq. (2.149), taking into account that $\vec{j}$ is antiparallel to $\vec{v}$, solving for $\vec{v}$, and inserting the resulting expression into eq. (2.153) gives

$$m_e^* \frac{d}{dt} \vec{E}(\vec{r}, t) + m_e^* \gamma \vec{E}(\vec{r}, t) = n_e \frac{e^2}{\sigma} \vec{E}(\vec{r}, t) . \quad (2.156)$$

Entering a plane wave ansatz for $\vec{E}$ according to eq. (2.9), differentiating and solving for σ yields

$$\sigma(\omega) = \frac{n_e e^2}{m_e^*} \frac{1}{\gamma - i\omega} . \quad (2.157)$$

When we enter this expression into eq. (2.152), we obtain

$$\tilde{n}^2 = \varepsilon_d - \frac{n_e e^2}{m_e^* \varepsilon_0} \frac{1}{\omega (\omega + i\gamma)} . \quad (2.158)$$

Here, the first term on the right side represents the contribution from the bound electrons and the second one that of the conduction (or free) electrons. The first contribution can be described by the Lorentz oscillator model and the second one by the Drude model. Their combination, which has led us to eq. (2.158), is known as the Drude–Lorentz model. We can simplify eq. (2.158) by introducing the plasma frequency

$$\omega_p := \sqrt{\frac{n_e e^2}{m_e^* \varepsilon_0 \varepsilon_d}} . \quad (2.159)$$

Then eq. (2.158) becomes

$$\tilde{n}^2 = \varepsilon_d \left(1 - \frac{\omega_p^2}{\omega (\omega + i\gamma)} \right) . \quad (2.160)$$

The meaning of this formula becomes evident when we evaluate it for different frequency ranges:

⁵ Here, we do not use $\vec{v}_d$ to describe the velocity of the electrons, since the dominating scattering processes at the higher frequencies of light are distinct from the ones in the electronic transport, which has been discussed in Chapter 1 of the book.

Case a) Very low frequencies: $\omega \ll \omega_p, \gamma$:

Equation (2.160) becomes

$$\tilde{n}^2 \approx \varepsilon_d \left(1 + \frac{i\omega_p^2}{\omega\gamma} \right). \quad (2.161)$$

In general, the damping rate γ is smaller than the plasma frequency ω_p , which implies

$$\tilde{n}^2 \approx \varepsilon_d \frac{i\omega_p^2}{\omega\gamma}. \quad (2.162)$$

The complex refractive index, hence, is

$$\tilde{n} \approx \sqrt{i} \sqrt{\varepsilon_d \frac{\omega_p^2}{\omega\gamma}} = \frac{1+i}{\sqrt{2}} \cdot \sqrt{\varepsilon_d \frac{\omega_p^2}{\omega\gamma}}. \quad (2.163)$$

At low frequencies, the bound electrons, in general, show no absorption, which implies that ε_d in eqs. (2.161), (2.162), and (2.163) is real. We obtain for the real refractive index and the extinction coefficient (eq. 2.22)

$$n, \kappa \approx \sqrt{\varepsilon_d \frac{\omega_p^2}{2\omega\gamma}}. \quad (2.164)$$

This means that at low frequencies, the values of n and κ are almost equal. Because κ is large due to $\omega \ll \omega_p, \gamma$, eq. (2.23) implies that the absorption coefficient α is large, too. Thus, metals absorb strongly at low frequencies. The penetration depth or skin depth of the wave,

$$\delta_{\text{skin}} := \frac{1}{\alpha}, \quad (2.165)$$

is then small (typically 20 nm to 50 nm at low frequencies). In other words, the light can penetrate only a very small distance into the metal. Because of the large values of κ and n , also the reflectivity R_γ is large (eq. (2.57)). For both reasons, a metal layer barely transmits light at low frequencies.

Case b) High frequencies below the plasma frequency: $\gamma \ll \omega < \omega_p$:

At high frequencies, eq. (2.160) becomes

$$\tilde{n} \approx \sqrt{\varepsilon_d \left(1 - \frac{\omega_p^2}{\omega^2} \right)}. \quad (2.166)$$

Provided that ε_d is real, eq. (2.166) implies that $\tilde{n}$ becomes purely imaginary. The real refractive index n then is zero, which means that the wave cannot propagate in the metal. Equation (2.57) implies that $R_\gamma \approx 1$, i.e., the metal surface is highly reflective. This gives metals their typical shine. However, the above condition that ε_d is real is not completely fulfilled for metals in the visible frequency range, due to the light absorption by bound electrons. Hence, the reflectivity drops below unity. For instance,

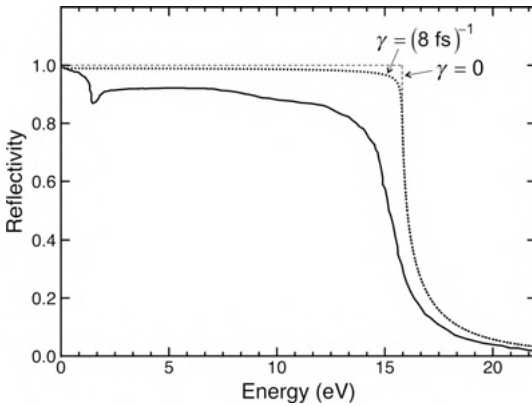


Fig. 2.35: Measured reflectivity spectrum of aluminium (solid line). For comparison, the dotted and dashed lines show spectra calculated using the free-electron model (eq. (2.158)) for $\epsilon_d = 1$, $\hbar\omega_p = 15.8$ eV and different damping rates γ . After Fig. 7.2 of [15], there after [22].

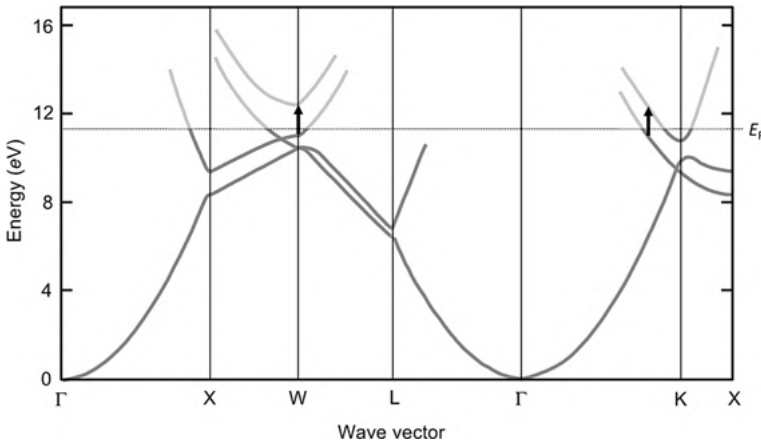


Fig. 2.36: Band structure of aluminium. States occupied with electrons are shaded dark grey, and empty states light grey. The dotted horizontal line marks the Fermi energy E_F . The interband transitions causing the reflection minimum at 1.5 eV in Fig. 2.35 are indicated by the arrows. After Fig. 7.3 of [15], there after [23].

the reflection spectrum of aluminium in Fig. 2.35 exhibits a minimum at 1.5 eV. At this photon energy, interband transitions between occupied states below the Fermi energy into empty states above it occur as indicated in Fig. 2.36 by arrows. In these positions, the marked pairs of bands run almost parallel to each other about 1.5 eV apart. This gives rise to a high joint density of states $g(\omega)$ at this photon energy and, hence, to a large absorption (eq. (2.110)). While the effect is smaller above 1.5 eV due to non-parallel band dispersions, R_γ is still considerably reduced there. The characteristic color differences of metals, such as aluminium, copper, silver, and gold, are hence

caused by differences between their interband transitions in the visible range, leading to a frequency dependent reflection.

Case c) High frequencies above the plasma frequency: $\omega > \omega_p$:

Equation (2.166) holds also in this case. In metals in which no transitions of bound electrons occur at these high frequencies, ϵ_d is real, so that $\tilde{n}$ is also real, resulting in a vanishing extinction coefficient κ (eq. 2.22). Indeed, metals such as the alkali metals and aluminium are non-absorbing for $\omega > \omega_p$. Their conduction electrons do not absorb because they have too much inertia to keep up with the frequent changes of direction of the electric field. Thus, they cannot take up energy from the light wave. According to eq. (2.57), the vanishing extinction coefficient for $\omega > \omega_p$ implies that at the plasma frequency, the reflectivity R_y will continuously drop from near unity to a smaller value determined by ϵ_d . An example of the reflectivity drop for aluminium is shown in Fig. 2.35 (solid line). For comparison, the results of an ideal free-electron model with and without damping (dotted and dashed line) are added. In the ideal metal case, ϵ_d is assumed to be unity so that R_y drops to zero. The fact that a similar drop is observed experimentally demonstrates that, near ω_p , the value of ϵ_d is indeed close to unity. The small or vanishing values of κ and R_y for $\omega > \omega_p$ imply that metals transmit light at ultraviolet wavelengths, an effect known as the ultraviolet transmission of metals.⁶

2.4.2 Particle plasmons

In the previous section, we have dealt with light reflection by extended metal surfaces. The high reflectivity of these surfaces gives metals their typical shine. The behavior of nanoparticles, i.e., particles with sizes between a few nm and ca. 100 nm, is very different, even if made of the same metals. As the extension of their surfaces is smaller than the wavelength of light, they do not reflect but scatter the light. This light can have brilliantly blue, green, or yellow colors, in contrast to the rather subtle colors of light reflected by extended metal surfaces. The colors of scattered light from metal nanoparticles are generated by collective oscillations of their conduction electrons relative to the positively charged ion background. These electron oscillations are excited by the alternating electric field of an incident light wave. An instantaneous picture of such an oscillation is shown in Fig. 2.37.

As the diameter of the nanoparticle is smaller than or similar to the penetration depth of the light, the alternating electric field of the light wave penetrates into the

⁶ Plasmons have also been observed in doped semiconductors, where the much lower charge carrier density n_e in the conduction or valence band leads to much lower plasma frequencies corresponding to $\hbar\omega_p \approx 0.1$ eV (eq. (2.159)) [24].

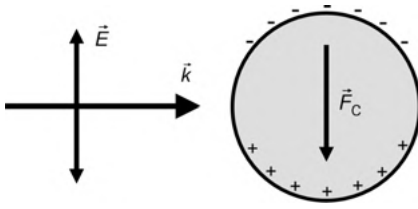


Fig. 2.37: Instantaneous distribution of polarization charges in a small spherical metal nanoparticle caused by the alternating electric field $\vec{E}$ of a light wave incident along $\vec{k}$. $\vec{F}_C$ is the restoring force exerted on the electrons by the positively charged ion cores.

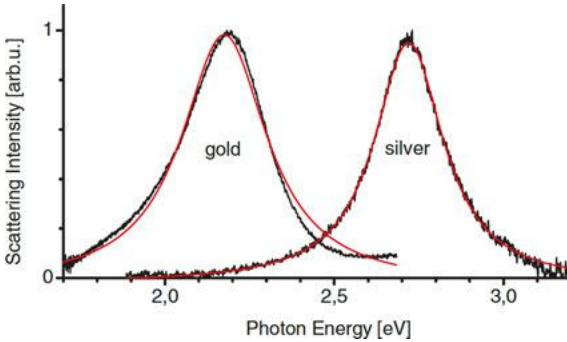


Fig. 2.38: The black curves show the measured light scattering spectra of two single nanoparticles made of gold and silver, respectively, as measured by microspectroscopy. The particles are spherical in shape and have a diameter of 60 nm. The red curves represent Lorentz fits to the data. Fig. 1 of [25].

metal and periodically displaces the conduction electrons. At the moment depicted in Fig. 2.37, the field has displaced all conduction electrons towards the top of the nanoparticle. This is followed by a displacement in the opposite direction in the next half cycle of the light field. The electron displacement results in polarization charges of opposite signs at the opposite surfaces of the nanoparticle and in a restoring force $\vec{F}_C$ exerted by the positive ion cores on the displaced electrons. The restoring force leads to an optical resonance, the particle plasmon resonance. Correspondingly, the energy quanta of the collective electron oscillation are called particle plasmons. Two examples of such resonances are shown in Fig. 2.38.

The spectral position of the resonance determines the color of the scattered light from the particles. This position is controlled by the size and shape of the particle as well as by the dielectric properties of the metal and its environment. The reason for these control parameters is that all these properties enter into the boundary conditions of the electromagnetic fields at the interface between the metal and its environment. Thus, they influence the force $\vec{F}_C$ exerted on the electrons. For instance, by solving the boundary problem, it is possible to show the following proportionality for the light power P_{scat} scattered by a small spherical metal nanoparticle of radius $\tilde{R}$:

$$P_{\text{scat}}(\omega) \propto \left| 4\pi\tilde{R}^3 \frac{\epsilon_m(\omega) - \epsilon_u(\omega)}{\epsilon_m(\omega) + 2\epsilon_u(\omega)} \right|^2. \quad (2.167)$$

Here, $\epsilon_m(\omega)$ is the dielectric function of the metal, and $\epsilon_u(\omega)$ is the dielectric function of the surrounding material. The resonance lies at the frequency ω at which the denominator in eq. (2.167) goes through a minimum. The line width of the resonance is determined by damping processes such as electron scattering, which are included in $\epsilon_m(\omega)$.

The unusual optical properties of metal nanoparticles and other metallic nanostructures have been investigated thoroughly over the last decades. This interest has been motivated by the wide range in which the spectral resonance positions of particle plasmons and related collective electron excitations can be tuned, and by the strong optical near-fields which are generated by the polarization charges at the metal surfaces and which are of interest for novel technical applications.

2.4.3 Reflection spectroscopy

A plot of the reflectivity versus the photon energy $\hbar\omega$ or wavelength λ of the light is called a reflection spectrum. Measuring such spectra and the methods applied in the measurement are known as reflection spectroscopy. Similarly to absorption spectra, reflection spectra can give valuable information about optical and electronic properties of a solid. Reflection spectroscopy can be especially helpful when the sample has a transmittance too small for absorption measurements. Figure 2.39 shows a typical experimental setup for reflection spectroscopy. Light from a white light source is spectrally filtered using a monochromator, and is incident on the sample. The reflected light power is measured with a detector, while the wavelength is varied using the monochromator. The detector should be positioned so that the angle between the incident and the reflected light is as small as possible to achieve nearly perpendicular incidence. To determine the reflectivity R_y from the reflected light power, often a comparison with the power reflected by a calibrated mirror is performed, instead of a normalization to the incident light power as in eq. (2.53). This approach avoids an interruption of the incident light beam to measure the incident power. An example of a measured reflection spectrum is shown in Fig. 2.35. An important variant of the re-

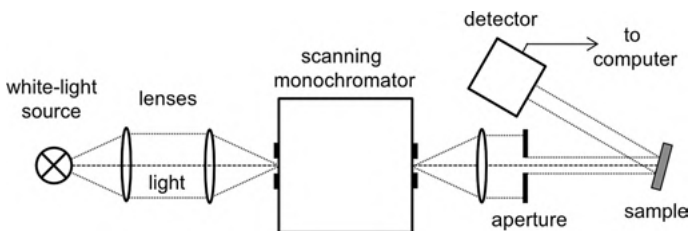


Fig. 2.39: Experimental set-up for the measurement of reflection spectra. Light from a white-light source is spectrally filtered with a monochromator and incident on the sample. The reflected light power is measured with a detector, while the wavelength is varied using the monochromator.

flection spectroscopy is ellipsometry, which is used to determine dielectric properties such as $\tilde{n}(\omega)$ and the thicknesses of thin layers. Its principle is based on determining the change of the polarization state of light upon reflection by a sample. For this purpose, light is obliquely incident on the sample with a polarization that is neither perpendicular nor parallel to the plane of incidence. The polarization properties of the reflected light are measured. As they depend on the optical constants and the thicknesses of the thin layers, both aspects can be determined.

2.5 Summary and outlook

This chapter of the book has treated the optical properties of solids. Microscopically, these properties result from the interaction of light with electrons in the medium. The goal has been to go into more depth on various microscopic aspects of the light-solid interaction, using quantum-mechanical and quantum-optical concepts when necessary and focusing on the important material classes of semiconductors and metals. We started off by reviewing some of the electrodynamics of light in solids. We then focused on a special kind of optical excitations in solids, namely interband transitions, in which an optical transition of an electron from one band to another occurs. They are of particular importance for the optical properties of semiconductors. Many optoelectronic devices are based on them, e.g., photodiodes, photovoltaic solar cells, light-emitting diodes, and semiconductor lasers. In particular, we have dealt with ways in which the electron and hole excited during an interband transition are bound to each other in a state called an exciton. We have seen how interband electron-hole excitations can, in turn, emit light, a light emission called luminescence. Subsequently, we have dealt with the free-electron currents in metals and doped semiconductors that give rise to their characteristic reflection spectra. Along the way, we have familiarized ourselves with some of the most common spectroscopic methods used to measure the optical properties of solids, namely absorption spectroscopy, photoluminescence spectroscopy and reflection spectroscopy.

The information provided here is regarded as a starting point to study more novel topics in optical research such as:

- quantization effects, which are based on electron confinement in semiconductor nanostructures (Section 1.3.1.1) and enable applications such as quantum-dot luminescence for color-television screens;
- nonlinear optics of solids, which is based on nonlinear responses at high light intensities and enables interesting applications such as optical frequency conversion;
- quantum-optical effects in solids, which are based on the quantum nature of the light field and give rise to phenomena such as photon entanglement;
- high-density excitation effects, which are based on many-body interactions in solids and can enable insights into different states of matter;

- optical metamaterials, which are based on sophisticated nanostructuring of optical materials and enable the manipulation of light in ways not available in nature, e.g., to fabricate optical cloaking devices.